

Report

Summer Project

Alireza Motazedian

Advisor:

Prof. MD Shaad Mahmud



University of
New Hampshire



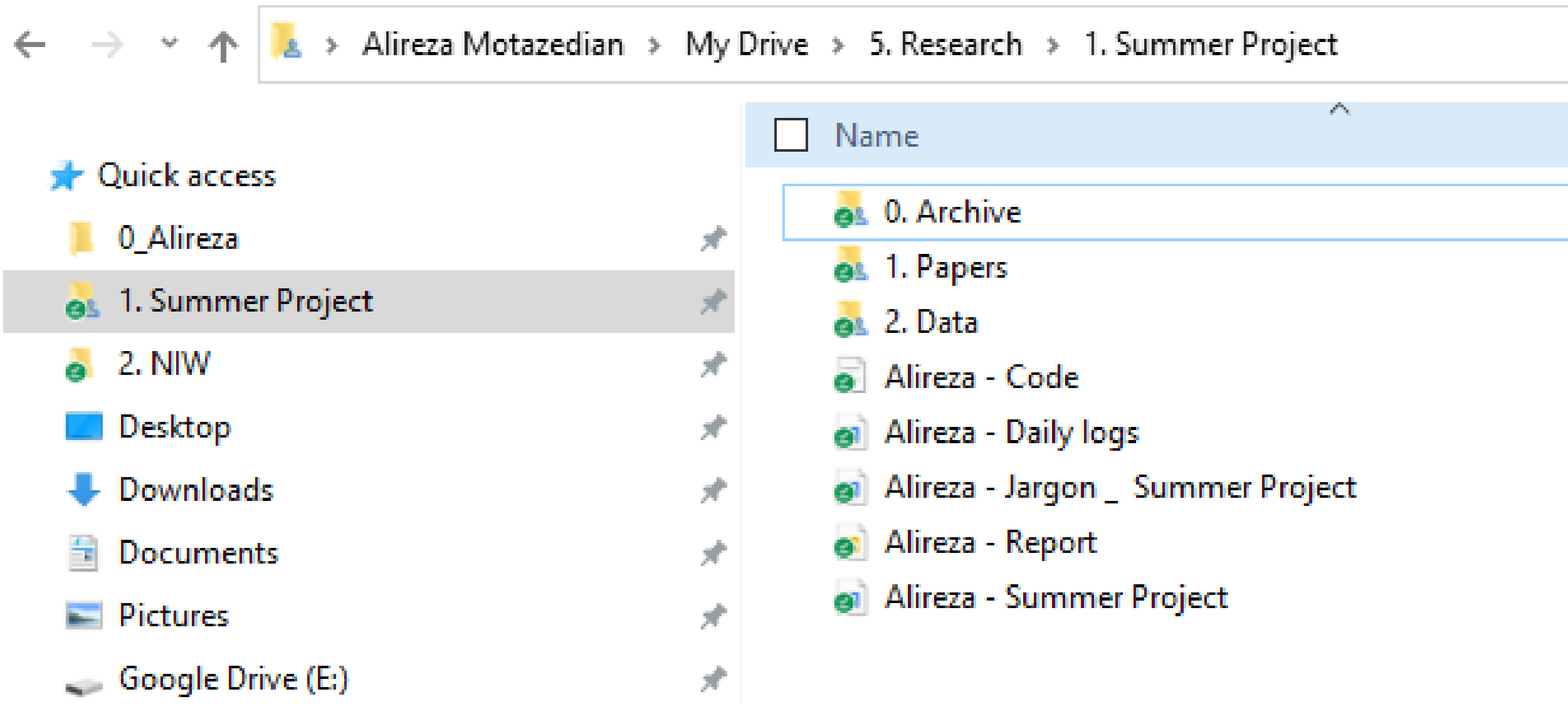
Weekly Report 1

05/25/2023

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- System Set-up and Files management
- Literature Review and Idea Generation
- Discussion with Talha
- Data Acquisition
- Need for Additional Suggestions



Weekly Report 2

06/01/2023

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CHAPTER

1

Societal and Economic Importance of Space Weather

Mike Hapgood

RAL Space, Harwell, Didcot, United Kingdom; Lancaster University, Lancaster, United Kingdom

CHAPTER OUTLINE

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1 WHAT IS SPACE WEATHER?

The past few decades have seen a growing awareness that phenomena in space can affect human activities on Earth. For example, this includes the impact of near-Earth objects, such as the small asteroid that exploded above Chelyabinsk in Russia that occurred on February 15, 2013. However, it also includes the range of phenomena that we call space weather, which

Comparison of Deep Learning Techniques to Model Connections Between Solar Wind and Ground Magnetic Perturbations

Amy M. Keesee^{1*}, Victor Pinto¹, Michael Coughlan¹, Connor Lennox², Md Shaad Mahmud² and Hyunju K. Connor³

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Specialty section:

Geomagnetically induced currents (GIC) can drive power outages and damage power grid components while also affecting pipelines and train systems. Developing the ability to predict local GICs is important to protecting infrastructure and limiting the impact of geomagnetic storms on public safety and the economy. While GIC data is not readily available, variations in the magnetic field, dB/dt , measured by ground magnetometers can be used as a proxy for GICs. We are developing a set of neural networks to predict the east and north components of the magnetic field, B_E and B_N , from which the horizontal component, B_H , and its variation in time, dB_H/dt , are calculated. We apply two techniques for time series analysis to study the connection of solar wind and interplanetary magnetic field properties obtained from the OMNI dataset to the ground magnetic field perturbations. The analysis techniques include a feed-forward artificial neural network (ANN) and a long-short term memory (LSTM) neural network. Here we present a comparison of both models' performance when predicting the B_H component of the Ottawa (OTT) ground magnetometer for the year 2011 and 2015 and then when attempting to reconstruct the time series of B_H for two geomagnetic storms that occurred on 5 August 2011 and 17 March 2015.

Revisiting the Ground Magnetic Field Perturbations Challenge: A Machine Learning Perspective

Victor A. Pinto^{1*}, Amy M. Keesee^{1,2}, Michael Coughlan², Raman Mukundan², Jeremiah W. Johnson³, Chigomezyo M. Ngwira⁴ and Hyunju K. Connor^{5,6}

¹Institute for the Study of Earth, Oceans and Space, University of New Hampshire, Durham, NH, United States, ²Department of Physics and Astronomy, University of New Hampshire, Durham, NH, United States, ³Department of Applied Engineering and Sciences, University of New Hampshire, Manchester, NH, United States, ⁴Orion Space Solutions, Louisville, CO, United States, ⁵NASA Goddard Space Flight Center, Greenbelt, MD, United States, ⁶Geophysical Institute, University of Alaska Fairbanks, Fairbanks, AK, United States

Forecasting ground magnetic field perturbations has been a long-standing goal of the space weather community. The availability of ground magnetic field data and its potential to be used in geomagnetically induced current studies, such as risk assessment, have resulted in several forecasting efforts over the past few decades. One particular community effort was the Geospace Environment Modeling (GEM) challenge of ground magnetic field perturbations that evaluated the predictive capacity of several empirical and first principles models at both mid- and high-latitudes in order to choose an operative model. In this work, we use three different deep learning models: a feed-forward neural network, a long short-term memory recurrent network and a convolutional neural network to forecast the horizontal component of the ground magnetic field rate of change (dB_H/dt) over 6 different ground magnetometer stations and to compare as directly as possible with

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A Review of Geomagnetically Induced Current Effects on Electrical Power System: Principles and Theory

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ABSTRACT Geomagnetically induced current (GIC) has been a significant concern for the electrical power grid in high latitudes for decades. Its origin starts in the Sun; during extreme space weather, the magnetic field of the Earth varies rapidly. This variation induces electric fields at the Earth's surface and leads to GICs in manmade technologies. Power systems are the most affected by this induced current, which causes half-cycle saturation of power transformers and other issues. Understanding the behaviours and chain effects of this phenomenon is the key consideration in modelling the hazards to technological systems from space weather. In this paper, a comprehensive review of space weather, geomagnetic disturbances (GMDs) and GICs and their impacts on the power systems in both high and mid-low latitude regions is presented. Additionally, we highlight the most commonly used methods to model and calculate geoelectric fields at the Earth's surface and GIC in the power systems with respect to DC and AC analysis. In addition, we have classified the GIC effects on the different power system components. Moreover, the possible solutions and mitigation techniques to eliminate or reduce these effects based on different GIC blocking devices are reviewed in this work. This work provides researchers and power system operators a shortcut road path to understanding GIC phenomena, modelling and calculations, effects, and mitigation of these effects.

Review

A Survey of Uncertainty Quantification in Machine Learning for Space Weather Prediction

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Abstract: With the availability of data and computational technologies in the modern world, machine learning (ML) has emerged as a preferred methodology for data analysis and prediction. While ML holds great promise, the results from such models are not fully reliable due to the challenges introduced by uncertainty. An ML model generates an optimal solution based on its training data. However, if the uncertainty in the data and the model parameters are not considered, such optimal solutions have a high risk of failure in actual world deployment. This paper surveys the different approaches used in ML to quantify uncertainty. The paper also exhibits the implications of quantifying uncertainty when using ML by performing two case studies with space physics in focus. The first case study consists of the classification of auroral images in predefined labels. In the second case study, the horizontal component of the perturbed magnetic field measured at the Earth's surface was predicted for the study of Geomagnetically Induced Currents (GICs) by training the model using time series data. In both cases, a Bayesian Neural Network (BNN) was trained to generate predictions, along with epistemic and aleatoric uncertainties. Finally, the pros and cons of both Gaussian Process Regression (GPR) models and Bayesian Deep Learning (DL) are weighed. The paper also provides recommendations for the models that need exploration, focusing on space weather prediction.

Keywords: artificial intelligence; uncertainty quantification; deep learning; machine learning; bayesian



Citation: Siddique, T., Mahmud, M.S., Keesee, A.M., Ngwira, C.M., Connor, H. A Survey of Uncertainty Quantification in Machine Learning for Space Weather Prediction. *Geosciences* 2022, 12, 27. <https://doi.org/10.3390/geosciences12010027>

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A Machine Learning Approach to Detection of Geomagnetically Induced Currents in Power Grids

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Abstract—Geomagnetically induced currents (GICs) in power grids are mainly caused by geomagnetic disturbances especially during solar storms. Such currents can potentially cause negative impacts on power grid equipment and even damage the power transformers resulting in a significant risk of blackouts. Therefore, monitoring GICs in power systems and developing solutions to mitigate their impacts before rising to a certain threatening level is urgently in need. Monitoring GICs is, however, quite a challenge and costly, as they usually appear in forms of dc components in the high voltage transmission lines, which are barely accessible through transformers. By examining the measured currents from the current transformers, this article proposes a framework to detect GICs in power transmission systems through a hybrid time-frequency analysis combined with machine learning technology. Simulated results verify that the proposed approach can promisingly estimate GICs in power systems during a variety of grid operating conditions.

Index Terms—Convolutional neural network (CNN), feature extraction, geomagnetic disturbance (GMD), geomagnetically induced current (GIC), harmonics, wavelet transform (WT).

I. INTRODUCTION

 GEOMAGNETIC disturbances (GMDs) are mainly caused

strong GMD caused major damages to electrical power equipment in Canada, Scandinavia, and the United States. Hydro-Quebec extra high voltage (EHV) transmission system experienced instability and tripping of lines carrying power to Montreal resulting in the total blackout of the Hydro-Quebec system [6], [7]. In the United States, a voltage fluctuation of up to 4% was recorded on the EHV systems in Pennsylvania, New Jersey, and Maryland. On September 19, 1989, a second solar storm damaged the step-up transformers at the Salem Unit 2 nuclear power plant [8].

Limiting the potential GICs-caused damages in power grids calls for developing advanced tools and mechanisms to monitor and detect GICs as they unfold and also solutions to mitigate the impacts before they rise to a certain threatening level. On one hand, the GMD phenomena may not always result in GICs in power systems. On the other hand, directly accessing GICs—which represents itself as a dc component in high voltage transmission lines—is costly and a challenge. Conventional techniques to monitoring high voltage transmission lines rely solely on the ac measurements through voltage transformers

Physics-Informed Machine Learning: A Survey on Problems, Methods and Applications

Zhongkai Hao, Songming Liu, Yichi Zhang, Chengyang Ying, Yao Feng,
Hang Su, Jun Zhu

Abstract—Recent advances of data-driven machine learning have revolutionized fields like computer vision, reinforcement learning, and many scientific and engineering domains. In many real-world and scientific problems, systems that generate data are governed by physical laws. Recent work shows that it provides potential benefits for machine learning models by incorporating the physical prior and collected data, which makes the intersection of machine learning and physics become a prevailing paradigm. By integrating the data and mathematical physics models seamlessly, it can guide the machine learning model towards solutions that are physically plausible, improving accuracy and efficiency even in uncertain and high-dimensional contexts. In this survey, we present this learning paradigm called Physics-Informed Machine Learning (PIML) which is to build a model that leverages empirical data and available physical prior knowledge to improve performance on a set of tasks that involve a physical mechanism. We systematically review the recent development of physics-informed machine learning from three perspectives of machine learning tasks, representation of physical prior, and methods for incorporating physical prior. We also propose several important open research problems based on the current trends in the field. We argue that encoding different forms of physical prior into model architectures, optimizers, inference algorithms, and significant domain-specific applications like inverse engineering design and robotic control is far from being fully explored in the field of physics-informed machine learning. We believe that the interdisciplinary research of physics-informed machine learning will significantly propel research progress, foster the creation of more effective machine learning models, and also offer invaluable assistance in addressing long-standing problems in related disciplines.

Index Terms—Physics-Informed Machine Learning, AI for Science, PDE/ODE, Symmetry, Intuitive Physics

Weekly Report 3

06/07/2023



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- Introduction :
- Problem Formulation :
- Possible Ways towards PIML :
- Applications of PIML :
- Conclusion :
- What we can learn from this paper about predicting GIC:

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Physics-Informed Machine Learning: A Survey on Problems, Methods and Applications

Zhongkai Hao, Songming Liu, Yichi Zhang, Chengyang Ying, Yao Feng, Hang Su, Jun Zhu

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Index Terms—Physics-Informed Machine Learning, AI for Science, PDE/ODE, Symmetry, Intuitive Physics

- ❖ Traditional machine learning algorithms have limitations in incorporating prior knowledge or constraints about physical systems .
- ❖ Physics-informed machine learning (PIML) is a new approach that seeks to integrate empirical data and physical prior knowledge into machine learning models .
- ❖ The paper presents a survey of recent developments in PIML and discusses its potential applications .

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- ❖ PIML addresses fundamental problems in physics-informed machine learning .
- ❖ Representation methodology for physical knowledge, approach for integrating physical knowledge into machine learning models, and practical problems that PIML resolves are discussed .
- ❖ The paper proposes a theoretical framework for machine learning problems with physical constraints based on probabilistic graphical models using latent variables to represent the real state of a system that satisfies physical prior constraints .
- ❖ The paper also discusses the challenges associated with incorporating physical prior knowledge into machine learning models such as the need for domain expertise, availability of data, and computational complexity .

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- ❖ The incorporation of physical prior knowledge can be achieved through modifications to one or more components of the machine learning model .
- ❖ Training data, model architecture, loss functions, optimization algorithms, and inference can all be modified to incorporate physical prior knowledge.
- ❖ The paper discusses several approaches for incorporating physical prior knowledge into machine learning models such as hybrid modeling, probabilistic graphical models, and feature engineering .
- ❖ The paper also provides examples of how different types of physical prior knowledge can be incorporated into machine learning models such as conservation laws, symmetries, and constitutive relations .

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- ❖ PIML has been successfully applied in various fields such as fluid dynamics, materials science, and robotics .
- ❖ By incorporating physical prior knowledge into machine learning models, improved performance and better alignment with practical problems that adhere to the laws of physics can be achieved .
- ❖ The paper provides examples of successful applications of PIML such as predicting fluid flow behavior in complex geometries, designing new materials with desired properties, and controlling robotic systems.

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- ❖ PIML is a promising new paradigm for solving physical problems using machine learning .
- ❖ By integrating empirical data and physical prior knowledge into machine learning models, it is possible to overcome the limitations of traditional machine learning algorithms and achieve improved accuracy and efficiency .
- ❖ The paper concludes by discussing future directions for research in PIML such as developing more efficient algorithms for incorporating physical prior knowledge, exploring new types of physical prior knowledge

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- ❖ The paper does not specifically address GIC prediction.
- ❖ The paper provides a theoretical framework for incorporating physical prior knowledge into machine learning models. This approach can be useful in predicting GIC by leveraging empirical data and available physical prior knowledge to improve performance on tasks that involve a physical mechanism.
- ❖ For example, the paper mentions that PIML has been applied in several topics of geography, including seismology, which involves the study of earthquakes and their effects on the Earth's magnetic field. Similarly, GIC is caused by variations in the Earth's magnetic field due to solar storms and other space weather events. By incorporating physical prior knowledge about the Earth's magnetic field and its interactions with solar wind and other factors, PIML can help improve predictions of GIC. This could involve modifying components of machine learning models such as training data, model architecture, loss functions, optimization algorithms, and inference to incorporate physical prior

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- ❖ My experience during my master's program
- ❖ Replicating a couple of your works as part of my summer project .

Comparison of Deep Learning Techniques to Model Connections Between Solar Wind and Ground Magnetic Perturbations

Amy M. Keese^{1*}, Victor Pinto¹, Michael Coughlan¹, Connor Lennox², Md Shaad Mahmud² and Hyunju K. Connor³

¹ Department of Physics & Astronomy and Space Science Center, University of New Hampshire, Durham, NH, United States,

² Department of Electrical and Computer Engineering, University of New Hampshire, Durham, NH, United States,

³ Department of Physics and Geophysical Institute, University of Alaska Fairbanks, Fairbanks, AL, United States

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Specialty section:

Geomagnetically induced currents (GIC) can drive power outages and damage power grid components while also affecting pipelines and train systems. Developing the ability to predict local GICs is important to protecting infrastructure and limiting the impact of geomagnetic storms on public safety and the economy. While GIC data is not readily available, variations in the magnetic field, dB/dt , measured by ground magnetometers can be used as a proxy for GICs. We are developing a set of neural networks to predict the east and north components of the magnetic field, B_E and B_N , from which the horizontal component, B_H , and its variation in time, dB_H/dt , are calculated. We apply two techniques for time series analysis to study the connection of solar wind and interplanetary magnetic field properties obtained from the OMNI dataset to the ground magnetic field perturbations. The analysis techniques include a feed-forward artificial neural network (ANN) and a long-short term memory (LSTM) neural network. Here we present a comparison of both models' performance when predicting the B_H component of the Ottawa (OTT) ground magnetometer for the year 2011 and 2015 and then when attempting to reconstruct the time series of B_H for two geomagnetic storms that occurred on 5 August 2011 and 17 March 2015.

Weekly Report 4

06/15/2023



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Weekly Report 5

06/22/2023



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Weekly Report 6

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Weekly Report 7

07/06/2023



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Unwatch 1

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<> Code



Alireza-Motazedian modified: README.md

69d652b 2 hours ago 19 commits

0. Archive	renamed Link	3 hours ago
1. Report	renamed daily to daily logs	3 hours ago
2. Code	update concept file: solar wind description	4 hours ago
3. Data	update code, data, ReadMe, Concepts, Daily logs, Reports	last week
4. Concepts _ Books _ Papers	deleted: paper links - modified: readme	2 hours ago
.gitignore	update .gitignore for desktop.ini	2 weeks ago
README.md	modified: README.md	2 hours ago
requirements.txt	init files	2 weeks ago

GIC

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Folder PATH listing
GIC:.
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|   README.md
|   requirements.txt
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|   Contract Summer 2023.pdf
|   Links _ Resources _ People.md
|   Proposal-GIC-Power Grid.pdf
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|
+---Books
|   01. Machine Learning Techniques for Space Weather.pdf
|
\---Papers
```

1. [Link](#) Replica _ team _ Keesee et al. 2022.pdf
2. [Link](#) Team _ Pinto et al. 2022.pdf
3. [Link](#) Team _ Physics-Informed Machine Learning A Survey.pdf
4. [Link](#) Team _ Analysis of Magnetometer Data from a Strong G3 Geomagnetic Disturbance.pdf
5. [Link](#) Team _ Real-Time Monitoring Applications for the Power Grid under Geomagnetic Disturbances.pdf
6. [Link](#) Team _ Sarker 2021.pdf
7. [Link](#) Team _ Siddique et al. 2022.pdf
8. [Link](#) Modelling considerations for the Hydro One real-time GMD management system.pdf
9. [Link](#) A Machine Learning Approach to Detection of Geomagnetically Induced Currents in Power Grids.pdf
10. [Link](#) A Review of Geomagnetically Induced Current Effects on Electrical Power System Principles and Theory.pdf
11. [Link](#) Deep Learning-Based Channel Prediction for Edge Co.pdf
12. [Link](#) Impact of geomagnetic disturbances on power transformers.pdf
13. [Link](#) Insight into impact of geomagnetically induced currents on power.pdf
14. [Link](#) Optimal Placement of GIC Blocking Devices Considering Equipment Thermal Limits and Power System Operation Constraints.pdf
15. [Link](#) Global Geomagnetic Perturbation Forecasting Using Deep Learning.pdf
16. [Link](#) The Role of GlobalRegional Earth Conductivity Models in Natural Geomagnetic Hazard Mitigation.pdf
17. [Link](#) Analysis of Geoeffective Impulsive Events on the Sun During the First Half of Solar Cycle 24.pdf
18. [Link](#) Measurement of geomagnetically induced current (GIC) around Tokyo, Japan.pdf


```
\---4. Concepts _ Books _ Papers
```

```
| Concepts.docx
```

```
+---Books
```

```
| 01. Machine Learning Techniques for Space Weather.pdf
```

```
\---Papers
```

1. [Link](#) Replica _ team _ Keesee et al. 2022.pdf
2. [Link](#) Team _ Pinto et al. 2022.pdf
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5. [Link](#) Team _ Real-Time Monitoring Applications for the Power Grid under Geomagnetic Disturbances.pdf
6. [Link](#) Team _ Sarker 2021.pdf
7. [Link](#) Team _ Siddique et al. 2022.pdf
8. [Link](#) Modelling considerations for the Hydro One real-time GMD management system.pdf
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10. [Link](#) A Review of Geomagnetically Induced Current Effects on Electrical Power System Principles and Theory.pdf
11. [Link](#) Deep Learning-Based Channel Prediction for Edge Co.pdf
12. [Link](#) Impact of geomagnetic disturbances on power transformers.pdf
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16. [Link](#) The Role of GlobalRegional Earth Conductivity Models in Natural Geomagnetic Hazard Mitigation.pdf
17. [Link](#) Analysis of Geoeffective Impulsive Events on the Sun During the First Half of Solar Cycle 24.pdf
18. [Link](#) Measurement of geomagnetically induced current (GIC) around Tokyo, Japan.pdf

Outlines

0. Jargon
1. Introduction
 - 1.1. Solar wind: What is it and how does it affect Earth?
 - 1.2. GIC
 - 1.3. Concept of UT
2. Website:
 - 2.1. SuperMAG
 - 2.1.1 SuperMAG
 - 2.1.2. Subtract Baseline
 - 2.1.3. High Fidelity and Low Fidelity
 - 2.2. Solar wind
3. Data Variable Definitions
 - 3.1. Solar wind (description)
 - 3.2. SuperMAG (description)
4. Machine Learning
 - 4.1. Time Series Analysis
 - 4.2. Kriging Techniques
 - 4.3. ADAM
5. Papers
 - 5.1. PILM: A Survey on Problems, Methods and Applications
 - 5.2. a real-time GMD monitoring system

GIC

Folder PATH listing

```
GIC:.  
| .gitignore  
| README.md  
| requirements.txt  
|  
+----.git  
|  
+----0. Archive  
|   Contract Summer 2023.pdf  
|   Links _ Resources _ People.md  
|   Proposal-GIC-Power Grid.pdf  
|  
+----1. Report  
|   Daily logs.md  
|   Report _ Slides.pptx  
|  
+----2. Code  
|   Data provided by Talha.ipynb  
|   GIC Data Exploration.ipynb  
|  
+----3. Data  
|   merged_1995-2011_2015.parq.zip  
|   README_SuperMAG_Year-by-year.txt  
|   solar_wind_1995-2022.parq.zip  
|   SuperMAG_1978-2021.parq.zip  
|   SuperMAG_1995.csv
```



Data provided by Talha.ipynb

File Edit View Insert Runtime Tools



Table of contents



0. Mount Google drive

1. Libraries



2. Data list



3. Data acquisition

3.1. Solar Wind (Omniweb)

3.2. SuperMAG

4. Data exploration

4.1. SuperMAG



Section

+---New_OTT (SuperMAG)

```
| | cleaned_OTT-2001-supermag.csv  
| | cleaned_OTT-2001-supermaghc.csv  
| | cleaned_OTT-2002-supermag.csv  
| | cleaned_OTT-2002-supermaghc.csv  
| | cleaned_OTT-2003-supermag.csv  
| | cleaned_OTT-2003-supermaghc.csv  
| | cleaned_OTT-2004-supermag.csv  
| | cleaned_OTT-2004-supermaghc.csv  
| | cleaned_OTT-2005-supermag.csv  
| | cleaned_OTT-2005-supermaghc.csv  
| | cleaned_OTT-2006-supermag.csv  
| | cleaned_OTT-2006-supermaghc.csv  
| | cleaned_OTT-2007-supermag.csv  
| | cleaned_OTT-2007-supermaghc.csv  
| | cleaned_OTT-2008-supermag.csv  
| | cleaned_OTT-2008-supermaghc.csv  
| | cleaned_OTT-2009-supermag.csv  
| | cleaned_OTT-2009-supermaghc.csv  
| | cleaned_OTT-2010-supermag.csv  
| | cleaned_OTT-2010-supermaghc.csv
```

\---Solar Wind (Omniweb)

```
solar_analyze.m  
solar_wind_2001.lst  
solar_wind_2002.lst  
solar_wind_2003.lst  
solar_wind_2004.lst  
solar_wind_2005.lst  
solar_wind_2006.lst  
solar_wind_2007.lst  
solar_wind_2008.lst  
solar_wind_2009.lst  
solar_wind_2010.lst  
solar_wind_2011.lst  
solar_wind_2012.lst  
solar_wind_2013.lst  
solar_wind_2014.lst  
solar_wind_2015.lst  
solar_wind_2016.lst  
solar_wind_2017.lst  
solar_wind_2018.lst
```

2. Code | GIC data exploration

5

GIC

Folder PATH listing

GIC:.

| .gitignore

| README.md

| requirements.txt

|

+---.git

|

+---0. Archive

| Contract Summer 2023.pdf

| Links _ Resources _ People.md

| Proposal-GIC-Power Grid.pdf

|

+---1. Report

| Daily logs.md

| Report _ Slides.pptx

|

+---2. Code

| Data provided by Talha.ipynb

| **GIC Data Exploration.ipynb**

|

+---3. Data

| merged_1995-2011_2015.parq.gzip

| README_SuperMAG_Year-by-year.txt

| solar_wind_1995-2022.parq.gzip

| SuperMAG_1978-2021.parq.gzip

| SuperMAG_1995.csv

GIC Data Exploration.ipynb ☆

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

Table of contents

- 1. Libraries & path
- 2. Data acquisition method
- 3. Data acquisition and compression
- 4. Merge datasets
- 5. Data exploration
- 5.4. Plots

1. Libraries & path

2. Data acquisition method

3. Data acquisition and compression

4. Merge datasets

5. Data exploration

5.1. Description

5.2. Response is dBe

5.3. Response is dBn

5.4. Plots

Next slide

2. Code | Data acquisition method (Thanks Talha)

6



lp All changes saved

+ Code + Text

1. Libraries & path

↳ 2 cells hidden

2. Data acquisition method

↳ 11 cells hidden

3. Data acquisition and compression

↳ 28 cells hidden

4. Merge datasets

↳ 6 cells hidden

5. Data exploration

5.1. Description

↳ 1 cell hidden

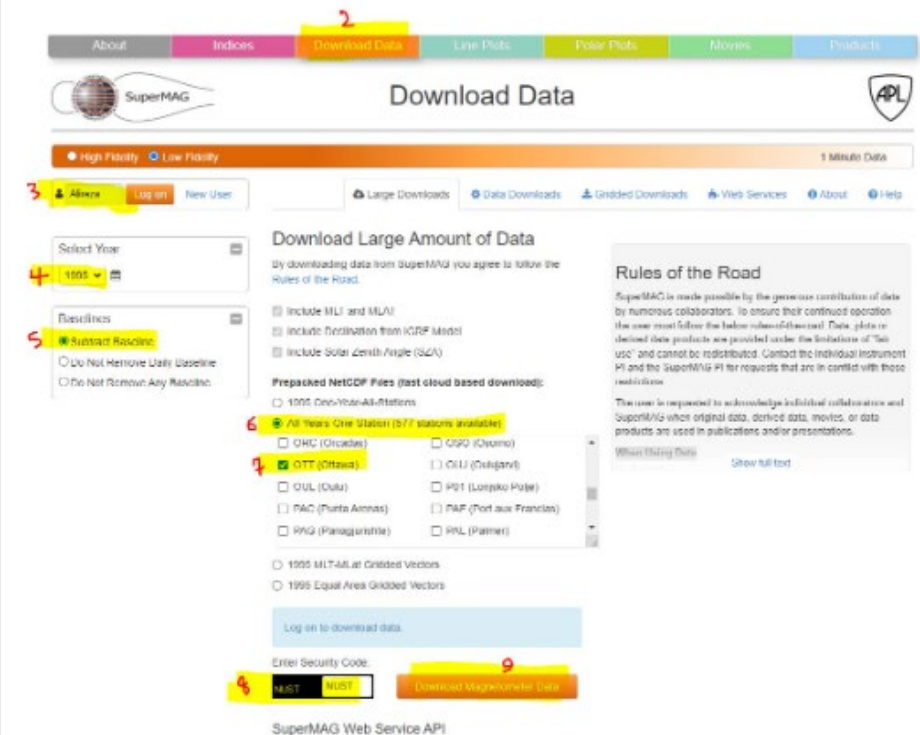
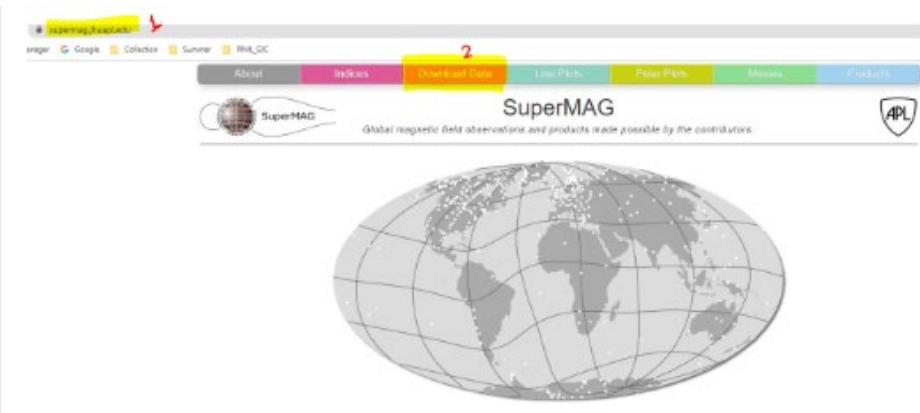
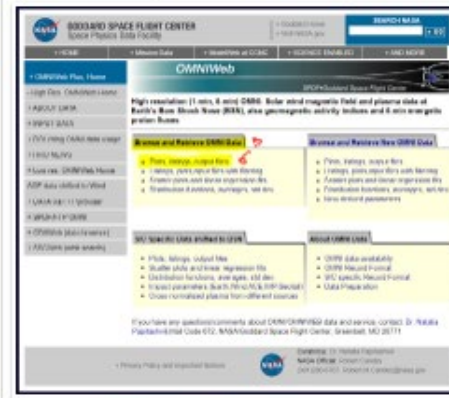
5.2. Response is dB

↳ 1 cell hidden

5.3. Response is dBn

↳ 1 cell hidden

5.4. Plots



3. Data | Solar wind data

7

Name	Owner	Last mo...	File size
 solar_wind_1995-2022.parq.gz 	 me	Jun 19, 2023	296.7 MB



```
3 urls = {
4   1995:"http://omniweb.gsfc.nasa.gov/staging/omni_min_WPz8o80Je1.lst",
5   1996:"http://omniweb.gsfc.nasa.gov/staging/omni_min_mWAY1KGXB.lst",
6   1997:"http://omniweb.gsfc.nasa.gov/staging/omni_min_bRTIvM2tjl.lst",
7   1998:"http://omniweb.gsfc.nasa.gov/staging/omni_min_QjDfhgPAk1.lst",
8   1999:"http://omniweb.gsfc.nasa.gov/staging/omni_min_WU162Lm_z7.lst",
9   2000:"http://omniweb.gsfc.nasa.gov/staging/omni_min_F4C29n8rDa.lst",
10  2001:"http://omniweb.gsfc.nasa.gov/staging/omni_min_DsD1ZtYvm7.lst",
11  2002:"http://omniweb.gsfc.nasa.gov/staging/omni_min_FEqatAmX1n.lst",
12  2003:"http://omniweb.gsfc.nasa.gov/staging/omni_min_b5LmphkKnK.lst",
13  2004:"http://omniweb.gsfc.nasa.gov/staging/omni_min_x4QrAEJ70Z.lst",
14  2005:"http://omniweb.gsfc.nasa.gov/staging/omni_min_NgK2TpXeEG.lst",
15  2006:"http://omniweb.gsfc.nasa.gov/staging/omni_min_1oLTQd5ThC.lst",
16  2007:"http://omniweb.gsfc.nasa.gov/staging/omni_min_hhw9ltigni.lst",
17  2008:"http://omniweb.gsfc.nasa.gov/staging/omni_min_cR0D1MKaTi.lst",
18  2009:"http://omniweb.gsfc.nasa.gov/staging/omni_min_J57Sx8m8RN.lst",
19  2010:"http://omniweb.gsfc.nasa.gov/staging/omni_min_imJhAjMVdI.lst",
20  2011:"http://omniweb.gsfc.nasa.gov/staging/omni_min_nDVNkiBfBP.lst",
21  2012:"http://omniweb.gsfc.nasa.gov/staging/omni_min_JAiF_lFZQ_.lst",
22  2013:"http://omniweb.gsfc.nasa.gov/staging/omni_min_8m7A_lPmNf.lst",
23  2014:"http://omniweb.gsfc.nasa.gov/staging/omni_min_RZh4xnWwX9.lst",
24  2015:"http://omniweb.gsfc.nasa.gov/staging/omni_min_WHZJchxh45.lst",
25  2016:"http://omniweb.gsfc.nasa.gov/staging/omni_min_o2KQdCUi1M.lst",
26  2017:"http://omniweb.gsfc.nasa.gov/staging/omni_min_XIcTeTcxMO.lst",
27  2018:"http://omniweb.gsfc.nasa.gov/staging/omni_min_1NsIci9LCc.lst",
28  2019:"http://omniweb.gsfc.nasa.gov/staging/omni_min_ISba1SdaZC.lst",
29  2020:"http://omniweb.gsfc.nasa.gov/staging/omni_min_w155smJ855.lst",
30  2021:"http://omniweb.gsfc.nasa.gov/staging/omni_min_sqdqxKg9zP.lst",
31  2022:"http://omniweb.gsfc.nasa.gov/staging/omni_min_z4x2sMrH2g.lst",
32 }
```

ITEMS	FORMAT
1 Year	I4
2 Day	I4
3 Hour	I3
4 Minute	I3
5 Field magnitude average, nT	F8.2
6 BX, nT (GSE, GSM)	F8.2
7 BY, nT (GSE)	F8.2
8 BZ, nT (GSE)	F8.2
9 Speed, km/s	F8.1
10 Vx Velocity, km/s	F8.1
11 Vy Velocity, km/s	F8.1
12 Vz Velocity, km/s	F8.1
13 Proton Density, n/cc	F7.2
14 Proton Temperature, K	F9.0
15 Flow pressure, nPa	F6.2
16 Electric field, mV/m	F7.2
17 SYM/H, nT	I6

Name	Owner	Last modified	File size
 SuperMAG_all_yeras.netcdf	 me	Jun 17, 2023 me	2.97 GB
 SuperMAG_1978-2021.parq.gzip	 me	8:34 PM me	758.1 MB

```
[ ] 1 df_SuperMAG.isna().sum()
```

```
date_time      0
dbe_geo        214340
dbe_nez        214340
dbn_geo        214340
dbn_nez        214340
dbz_geo        214340
dbz_nez        214340
decl           0
mcolat         0
mlat           0
mlon           0
mlt            0
sza            0
dtype: int64
```

```
[ ] 1 # The number of days with missing data
2 # min=60, hour=24,
3 days = 214340/(60*24)
4 print(days)
5
6 # The percentage of missing data
7 round(214340 / 23142240, 3)
```

```
148.84722222222223
```

Dealing with missing values:

As the data is time-series data, it would be better not to drop missing values. Instead, we can fill NAs with median or miss forest.

- Large downloads: all years
- Data downloads: year by year

Login Log on New User

Large Downloads

Data Downloads

Select Year

2001



Baselines

☒ Subtract Baseline

☐ Do Not Remove Daily Baseline

☐ Do Not Remove Any Baseline

Download Large Amount of Data

By downloading data from SuperMAG you agree to follow the [Rules of the Road](#).

☒ Include MLT and MLAT

☒ Include Declination from IGRF Model

☒ Include Solar Zenith Angle (SZA)

Prepacked NetCDF Files (fast cloud based download):

☐ 2001 One-Year-All-Stations

☒ All-Years-One-Station (577 stations available)

```
'df_SuperMAG_all_years_1995:'  
(525600, 13)
```

	date_time	dbe_geo	dbe_nez	dbn_geo	dbn_nez	dbz_geo	dbz_nez	decl	mcolat	mlat	mlon	mlt	sza
0	1995-01-01T00:00:00	-1.7	-0.7	3.6	3.9	4.8	4.8	-14.41	33.67	56.33	0.75	18.89	115.55
1	1995-01-01T00:01:00	-1.7	-0.7	3.6	3.9	4.8	4.8	-14.41	33.67	56.33	0.75	18.91	115.73

```
'df_SuperMAG_Year_by_year_1995:'  
(525600, 13)
```

	date_time	dbe_geo	dbe_nez	dbn_geo	dbn_nez	dbz_geo	dbz_nez	decl	mcolat	mlat	mlon	mlt	sza
0	1995-01-01 00:00:00	-1.656206	-1.656206	3.560276	3.860474	4.760194	4.760194	-14.413527	33.673729	56.326271	0.749585	18.892572	115.552635
1	1995-01-01 00:01:00	-1.681890	-1.681890	3.577185	3.883245	4.756423	4.756423	-14.413527	33.673729	56.326271	0.749586	18.909193	115.726540

```
'df_SuperMAG_Year_by_year_1995:'  
(525600, 17)
```

	Date.UTC	Extent	IAGA	GEOLON	GEOLAT	MAGON	MAGLAT	MLT	MCOLAT	IGRF_DECL	SAZ	dbn_nez	dbe_nez	dbz_nez	dbn_geo	dbe_geo	dbz_geo
0	1995-01-01T00:00:00	60	OTT	284.45	45.4	0.75	56.33	18.89	33.67	-14.41	115.55	3.9	-0.7	4.8	3.6	-1.7	4.8
1	1995-01-01T00:01:00	60	OTT	284.45	45.4	0.75	56.33	18.91	33.67	-14.41	115.73	3.9	-0.7	4.8	3.6	-1.7	4.8

By My search

Here are the descriptions for each column in your "SuperMAG_1995.csv" dataset:

Date.UTC: This represents the timestamp of the observation in Coordinated Universal Time (UTC).

Extent: This could potentially represent the extent of a particular magnetic event, but without specific documentation, it's hard to say exactly. It could also be a geographic or magnetic extent, such as the coverage of a magnetometer station.

IAGA: This stands for International Association of Geomagnetism and Aeronomy. The IAGA code is a unique identifier given to each magnetometer station.

GEOLON (Geographic Longitude): This is the geographic longitude of the magnetometer station, measured in degrees from -180 to +180.

GEOLAT (Geographic Latitude): This is the geographic latitude of the magnetometer station, measured in degrees from -90 (South Pole) to +90 (North Pole).

MAGON (Magnetic Longitude): This is the longitude of the magnetometer station in geomagnetic coordinates.

MAGLAT (Magnetic Latitude): This is the latitude of the magnetometer station in geomagnetic coordinates.

MLT (Magnetic Local Time): This is a measure of solar time based on the location's magnetic longitude.

MCOLAT (Magnetic Co-latitude): This is 90 degrees minus the magnetic latitude, measuring the angle from the magnetic north pole.

IGRF_DECL (International Geomagnetic Reference Field Declination): This is the angle between magnetic north and true north at a particular location, according to the International Geomagnetic Reference Field model.

SAZ (Solar Zenith Angle): This is the angle between the line perpendicular to the Earth's surface at a given location and the line from that location to the Sun. It can be used to infer the level of solar radiation reaching the location.

dbn_nez, dbe_nez, dbz_nez: These represent the changes (delta, denoted by 'db') in the northward, eastward, and downward components of the magnetic field in NEZ (North, East, Down) coordinates.

dbn_geo, dbe_geo, dbz_geo: These represent the changes (delta, denoted by 'db') in the northward, eastward, and downward components of the magnetic field in geographic coordinates.

High Fidelity vs. Low Fidelity | Question?

11

[About](#)[Indices](#)[Download Data](#)[Line Plots](#)[Polar Plots](#)[Movies](#)[Products](#)



Download Data



☒ High Fidelity ☐ Low Fidelity 1 Second Data

 Alireza Motazedian [Change](#)
[Log out](#)

[Large Downloads](#)[Data Downloads](#)[ULF Parameters](#)[Web Services](#)[About](#)[Help](#)

Select Day
1 January 2001

Download Large Amount of Data

By downloading data from SuperMAG you agree to follow the [Rules of the Road](#).

☒ Include MLT and MLAT
☒ Include Declination from IGRF Model
☒ Include Solar Zenith Angle (SZA)
☐ Include ULF Integrated Power (Pc2 - Pc5)

Format:
☐ ASCII (self documented)
☐ CSV (comma separated)
☒ NetCDF

Download Duration:
☒ 2001-01-01 One-Day-All-Stations

Enter Security Code:
BJR1

[Download Magnetometer Data](#)

Rules of the Road

SuperMAG is made possible by the generous contribution of data by numerous collaborators. To ensure their continued operation the user must follow the below rules-of-the-road. Data, plots or derived data products are provided under the limitations of "fair use" and cannot be redistributed. Contact the individual instrument PI and the SuperMAG PI for requests that are in conflict with these restrictions.

The user is requested to acknowledge individual collaborators and SuperMAG when original data, derived data, movies, or data products are used in publications and/or presentations.

[When Using Data](#)[Show full text](#)

Thanks, Talha.

Answer:
Low Fidelity

Reason: 1-min

Comparison of Deep Learning Techniques to Model Connections Between Solar Wind and Ground Magnetic Perturbations

Amy M. Keesee^{1}, Victor Pinto¹, Michael Coughlan¹, Connor Lennox², Md Shaad Mahmud² and Hyunju K. Connor³*

- Improve the model including adjustments of input parameters.
- Increased time history cadence
- Comparison to additional time series techniques
- More ground station

Revisiting the Ground Magnetic Field Perturbations Challenge: A Machine Learning Perspective

Victor A. Pinto^{1}, Amy M. Keesee^{1,2}, Michael Coughlan², Raman Mukundan², Jeremiah W. Johnson³, Chigomezoyo M. Ngwira⁴ and Hyunju K. Connor^{5,6}*

- Hyper-parameter tuning process
- Real-time forecasting

- Hyper-parameter tuning process
- ARIMA
- Other DL models



merged_1995-2011_2015.parq.gzip



me

Jul 3, 2023

289.7 MB

'dfe:'
(9466560, 15)

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dBe
0	1995-01-01 00:00:00	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.760194	-1.656206
1	1995-01-01 00:01:00	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.756423	-1.681890

'dfn:'
(9466560, 15)

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dBn
0	1995-01-01 00:00:00	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.760194	3.860474
1	1995-01-01 00:01:00	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.756423	3.883245

Weekly Report 8

07/13/2023

UNH
ECE

DEPT.

- multi-class classifications
- multi-label classifications

$$\left(\frac{dB}{dt}\right)_H = \sqrt{\frac{dB_N^2}{dt} + \frac{dB_E^2}{dt}}$$

where

$$\frac{dB_i}{dt} = \frac{[B_i(t + 1 \text{ min}) - B_i(t)]}{1 \text{ min}}$$

Stage: merged_1995-2011_2015.parq.gzip
(9466560, 16)

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dB _e	dB _n	
0	1995-01-01	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.760194	-1.656206	3.860474	

Stage: dBH = sqrt(dBe**2 + dBn**2)
(9466560, 17)

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dB _e	dB _n	dBH
0	1995-01-01	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.760194	-1.656206	3.860474	4.200747

Stage: dropped: dB_e & dB_n
(9466560, 15)

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dBH
0	1995-01-01	0.19	0.9	-0.98	1.35	99999.9	99999.9	99999.9	99999.9	999.99	9999999.0	99.99	999.99	4.760194	4.200747

```
[ ] 1 dfH.isna().sum()
```

```
date_time      0
Bx              0
By              0
Bz              0
BT              0
Vx              0
Vy              0
Vz              0
VT              0
n_p             0
T_p             0
P_p             0
E               0
dBz            97333
dBH            97333
dtype: int64
```

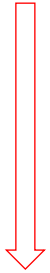
Dealing with missing values:

- Drop
- Fill
 - Mean
 - Median
 - Mode
 - Constant number or string, like 0 or 'unknown'
 - Mice Forest
 - Miss Forest
 - KNN Imputation

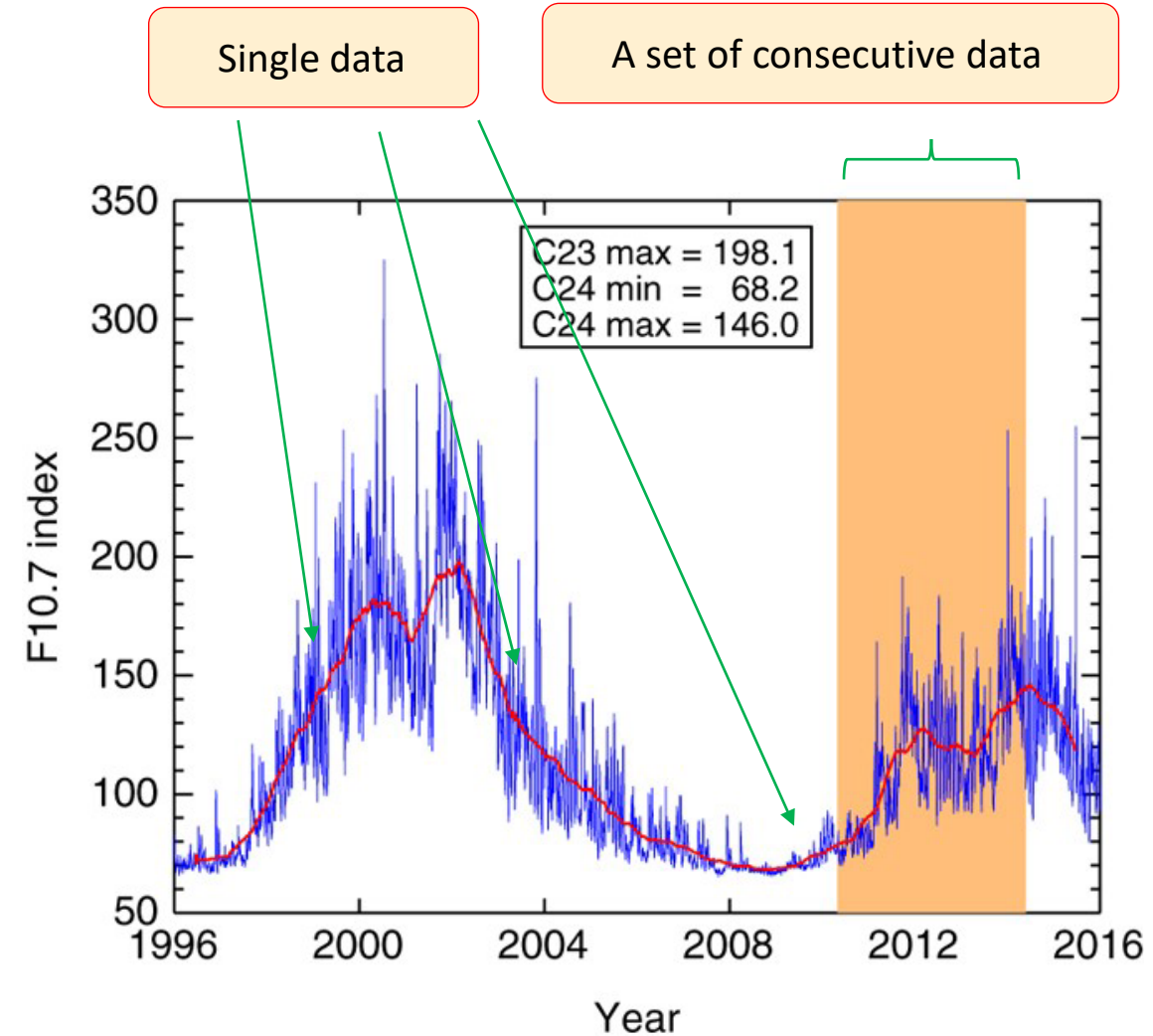
Dropping missing values: rejected

3

1. Dropping missing values



Not efficient in our work



- Fill
 - Mean
 - Median
 - Mode
 - Constant number or string, like 0 or 'unknown'
 - Miss Forest
 - Mice Forest
 - KNN Imputation

Miss Forest

- **well-suited** for datasets where the missing values occur in a **pattern** rather than randomly.
 - based on the Random Forest model
 - The basic idea of the Miss Forest algorithm is to treat each feature with missing values as the dependent variable and the other features as the independent variables. A Random Forest model is trained for each feature with missing values using the other features. The missing values are then predicted based on these trained models.

It's important to **remember** that:

- imputing missing values **always** comes with some **uncertainty**.
- It's **generally** a good idea to explore your data thoroughly and understand the reasons behind missing values before deciding on an imputation strategy.
- Also, while Miss Forest is a powerful imputation method, it may not be suitable for all kinds of data or missing patterns, and it **may be computationally expensive** for large datasets.

MICE:

- **MI**s (**M**ultiple **I**mutations) create **several** possible fills for each missing value
 - The **distribution**: a good approximation
 - **Flexible** (can handle different data types)
- **RF** (**R**andom **F**orest)
 - **Powerful**
- **MICE = MI**s + **RF**
 - MICE **combines** the flexibility of MICE with the power of random forests. This approach can handle different data types (like categorical and numerical values) and missing data patterns.
 - **MICE** (**M**ultivariate **I**mutation by **C**hained **E**quations)
 - is an imputation method that works by filling the missing data **multiple times**.

- KNN (K-Nearest Neighbors)
- The **optimal K** may depend on the specific dataset and problem.
- **Can be time-consuming** on larger datasets as each missing value needs to compute distances to all other points.

Assumption: KNN assumes that the dataset has a metric space structure, which may not be the case for all datasets.

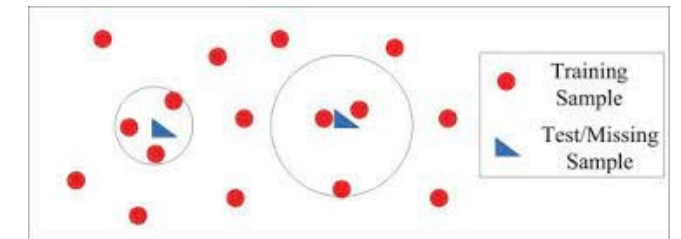
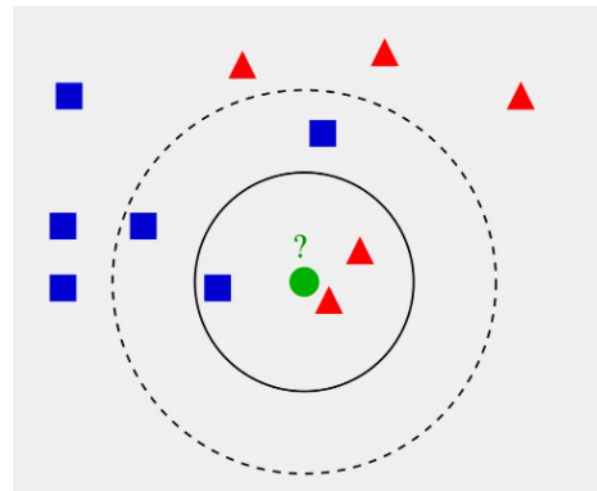
Euclidean (metric) distance
vs
Manhattan (taxi-cab) distance



Classification
Majority vote

KNN

Regression
Mean



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```

1995 1 1 2 3.22 -0.40 2.00 -2.49 313.1 -312.0 0.2 26.1 17.43 19271. 3.42 0.60 11
1995 1 1 3 3.19 -0.63 2.48 -1.86 312.6 -311.9 -0.5 20.8 17.38 18251. 3.40 0.37 12
1995 1 1 4 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.99 99.99 999.99 12
1995 1 1 5 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.9 99.99 999.99 12
1995 1 1 6 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.9 99.99 999.99 11
1995 1 1 7 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.9 99.99 999.99 11
1995 1 1 8 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.9 99.99 999.99 11
1995 1 1 9 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.9 99.99 999.99 11
1995 1 1 10 9999.99 9999.99 9999.99 9999.99 9999.9 9999.9 9999.9 9999.9 99.99 999.99 11
1995 1 1 11 3.26 -0.57 2.59 -1.89 313.5 -312.8 0.0 20.2 17.22 15199. 3.38 0.37 12
1995 1 1 12 3.15 -0.53 2.48 -1.85 313.5 -312.8 0.0 20.2 17.22 15199. 3.38 0.37 12
1995 1 1 13 3.05 -0.70 2.61 -1.42 314.4 -313.7 -0.5 21.4 17.05 16473. 3.37 0.23 12
1995 1 1 14 3.05 -0.70 2.47 -1.65 9999.9 9999.9 9999.9 9999.9 99.99 999.99 12
1995 1 1 15 3.05 -0.69 2.40 -1.75 313.4 -312.6 0.0 23.0 17.96 17568. 3.53 0.35 12
1995 1 1 16 3.03 -0.53 2.34 -1.85 313.4 -312.6 0.0 23.0 17.96 17567. 3.53 0.38 12
1995 1 1 17 3.07 -0.66 2.55 -1.55 313.1 -312.4 1.1 21.3 18.13 17126. 3.55 0.28 13
1995 1 1 18 3.05 -0.91 2.65 -1.20 314.2 -312.7 1.6 30.1 18.21 14372. 3.60 0.16 12
1995 1 1 19 3.11 -0.80 2.76 -1.18 315.9 -314.2 -5.5 32.5 18.46 16909. 3.68 0.15 12
1995 1 1 20 3.13 -0.52 2.32 -2.02 314.6 -313.1 0.5 30.7 17.55 19147. 3.47 0.44 12
1995 1 1 21 3.00 -0.39 2.28 -1.89 314.6 -313.1 0.5 30.7 17.55 19147. 3.47 0.40 12
1995 1 1 22 3.07 -0.28 2.28 -2.05 9999.9 9999.9 9999.9 9999.9 99.99 999.99 12
1995 1 1 23 2.89 -0.32 2.20 -1.85 313.1 -312.4 0.5 21.3 17.05 18238. 3.34 0.40 13
1995 1 1 24 3.17 -0.27 2.40 -2.04 9999.9 9999.9 9999.9 9999.9 99.99 999.99 13
1995 1 1 25 3.01 -0.26 2.23 -2.00 312.8 -311.9 0.0 24.0 17.22 17347. 3.37 0.44 13
1995 1 1 26 3.08 -1.00 2.63 -0.92 313.1 -312.4 1.6 21.3 18.46 15408. 3.62 0.08 13
1995 1 1 27 3.00 -1.09 2.64 -0.90 9999.9 9999.9 9999.9 9999.9 99.99 999.99 13
1995 1 1 28 3.07 -1.10 2.64 -1.06 9999.9 9999.9 9999.9 9999.9 99.99 999.99 13
1995 1 1 29 3.10 -0.87 2.54 -1.55 314.9 -313.4 1.1 30.2 18.54 14576. 3.68 0.28 13

```

streaks

Stage: df_NAs
(97333, 15)

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dBH
57681	1995-02-10 01:21:00	3.56	3.62	3.26	6.03	-349.4	-10.0	-0.8	349.6	7.55	9594.0	1.85	-1.6	NaN	NaN

The number of missing-streaks: 7803

The maximum length of missing-streaks: 9754

The minimum length of missing-streaks: 3

The average length of missing-streaks: 12

The Median length of missing-streaks: 7

The number of unique length of missing-streaks: 120

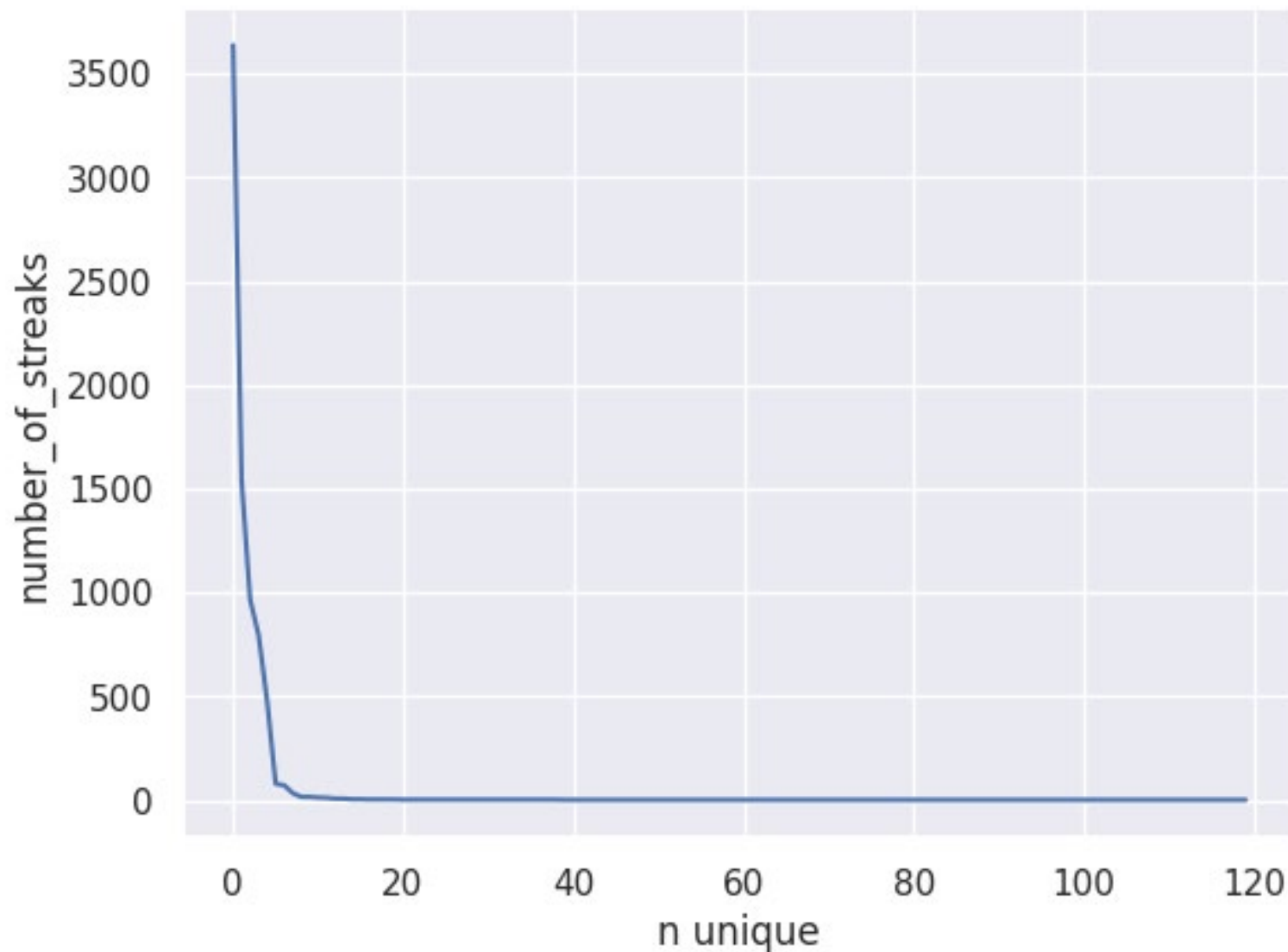
The number of unique value-count of length of missing-streaks: 16

The unique value-count of length of missing-streaks: [3637 1539 966 794 477 79 71 34 15 12 11 7 4 3
2 1]

```
'df_missing_streaks':  
(120, 2)
```

	unique_length	number_of_streaks
0	7	3637
1	6	1539
2	8	966
3	5	794
4	4	477
...	...	...
115	1481	1
116	145	1
117	57	1
118	105	1
119	531	1

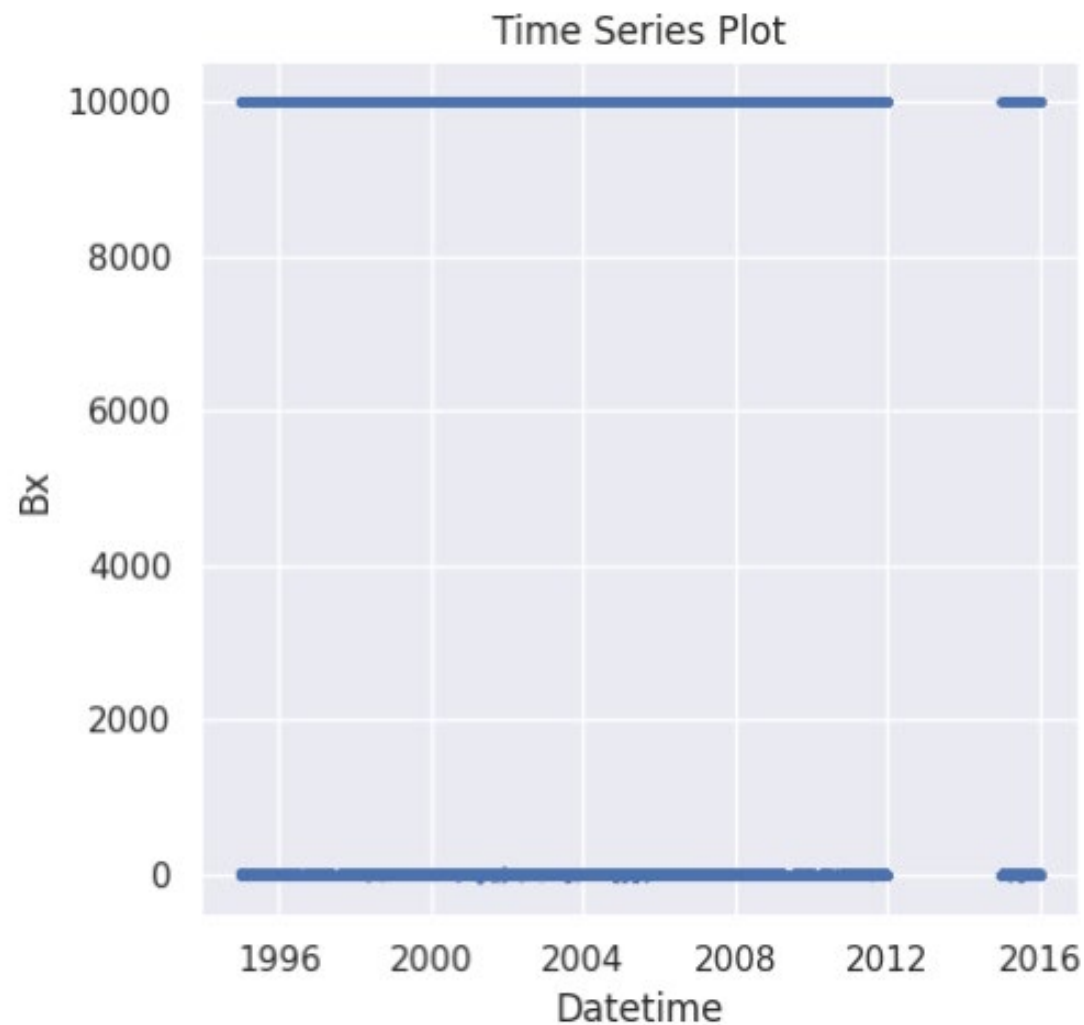
120 rows × 2 columns



	Bx	By	Bz	BT	Vx	Vy \
count	9466560.00	9466560.00	9466560.00	9466560.00	9466560.00	9466560.00
mean	695.33	695.42	695.33	700.87	18657.15	19009.15
std	2543.62	2543.59	2543.62	2542.10	39408.95	39238.31
min	-49.51	-48.05	-52.51	0.12	-1073.00	-864.90
25%	-2.56	-2.53	-1.58	4.04	-467.90	-12.10
50%	0.41	0.51	0.23	5.47	-387.20	2.70
75%	3.33	3.55	2.26	7.73	-314.20	34.20
max	9999.99	9999.99	9999.99	9999.99	99999.90	99999.90

	Vz	VT	n_p	T_p	P_p	E \
count	9466560.00	9466560.00	9466560.00	9466560.00	9466560.00	9466560.00
mean	19007.70	19363.71	195.31	1983264.86	20.76	194.54
std	39239.01	39066.64	389.88	3890703.21	38.36	395.83
min	-310.40	230.50	0.03	1833.00	0.01	-56.33
25%	-11.60	372.10	3.53	40707.00	1.36	-0.54
50%	2.70	445.00	6.05	90798.00	2.11	0.35
75%	27.20	613.60	13.94	232620.00	4.31	2.15
max	99999.90	99999.90	999.99	9999999.00	99.99	999.99

	dBz	dBH
count	9466560.00	9466560.00
mean	1.19	13.75
std	18.68	22.87
min	-1470.00	0.00
25%	-1.33	4.60
50%	1.44	8.68
75%	4.78	15.98
max	737.90	2122.18

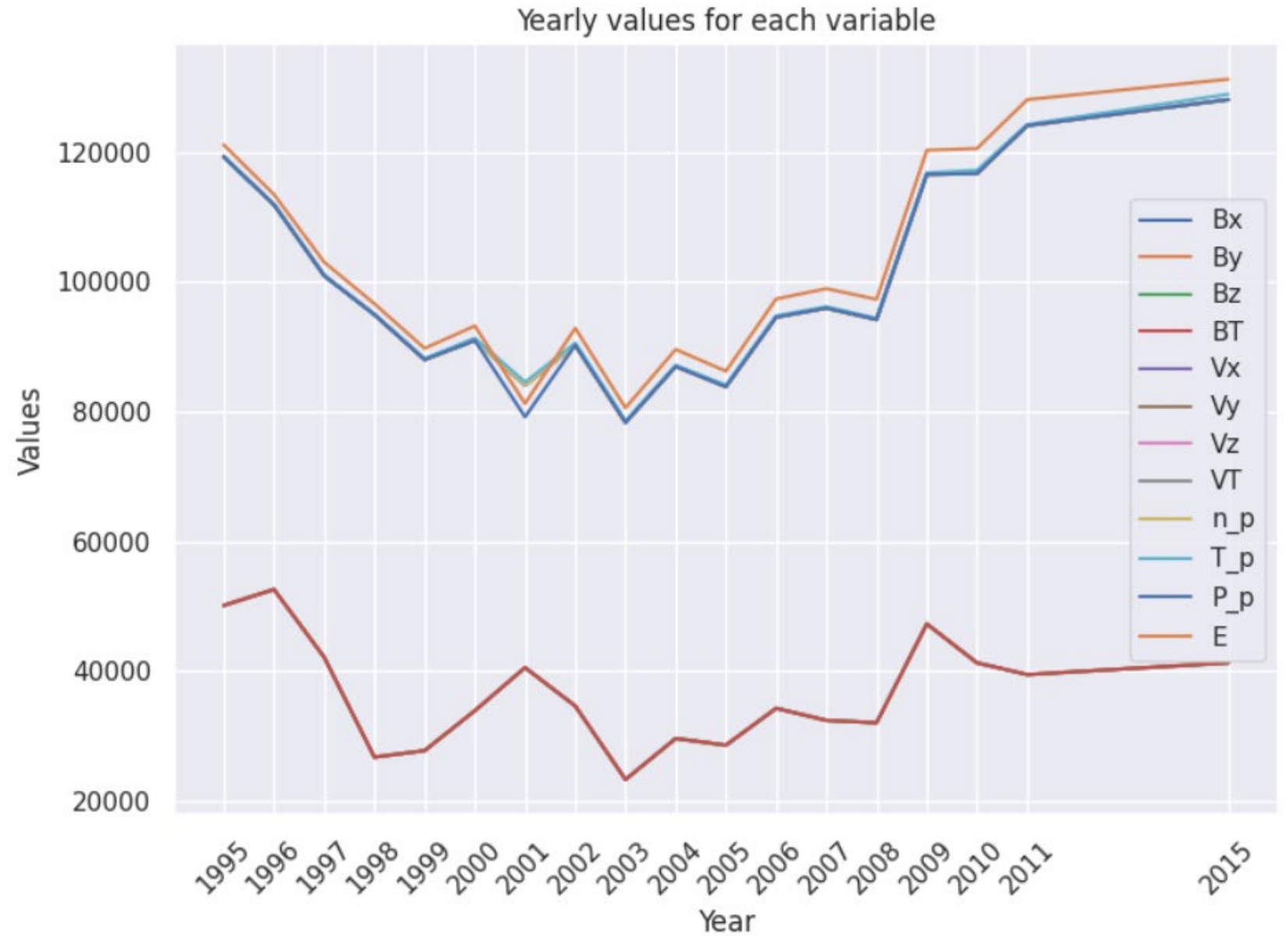


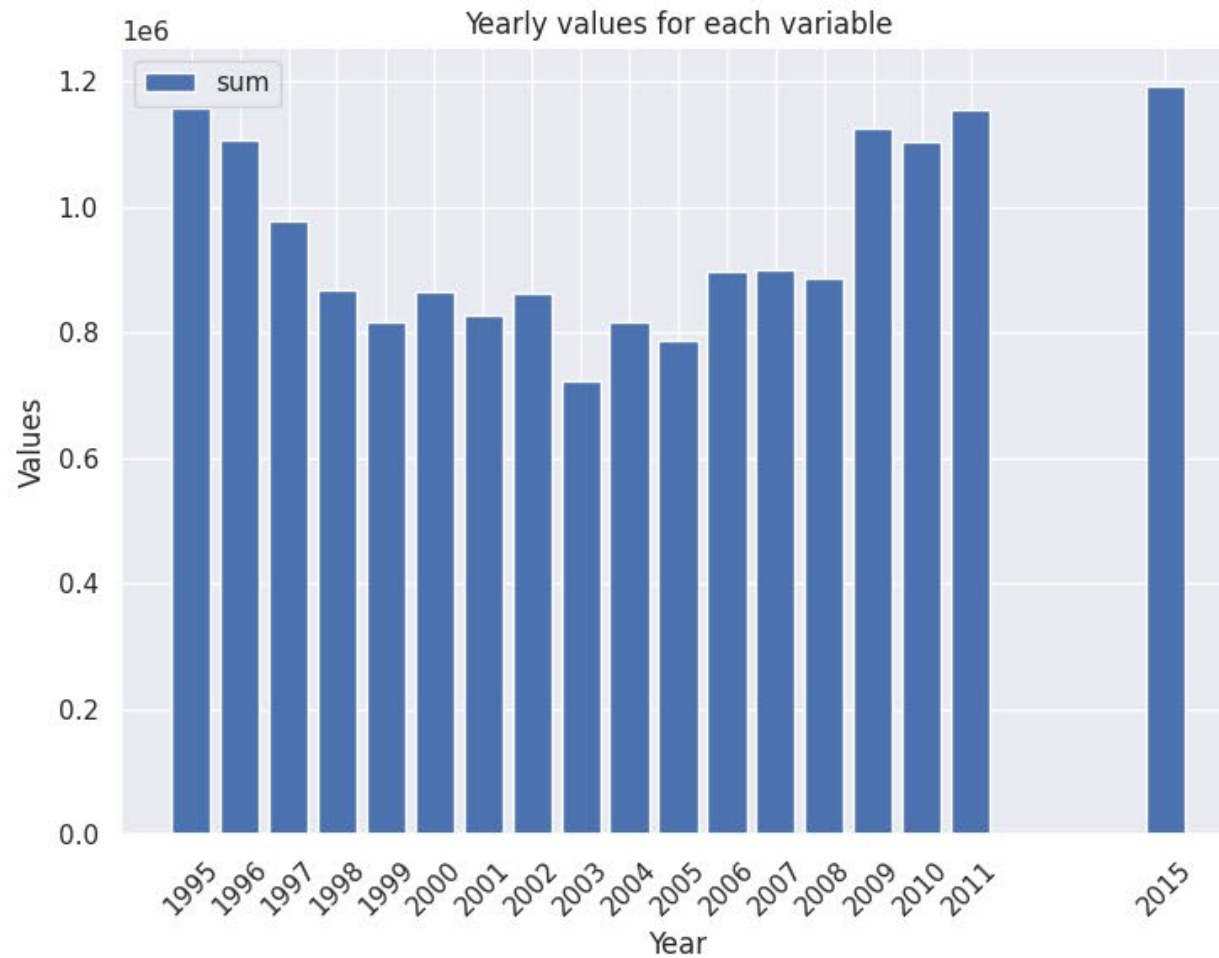
```
-----> ["Bx"]
      rows: 9,466,560
      rows with weird values: 658,255
      Years with weird values: 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2015
The number of unique counts: 5,540
value_____count 9999.99      658255
-2.54          9159
2.04           9065
-2.46          9061
-2.42          8985
Name: Bx, dtype: int64
```

```
'df_filled:'
(9466560, 15)
```

	date_time	Bx	By	Bz	BT	Vx	Vy	Vz	VT	n_p	T_p	P_p	E	dBz	dBH
0	1995-01-01 00:00:00	0.19	0.90	-0.98	1.35	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	4.76	4.20
1	1995-01-01 00:01:00	0.19	0.90	-0.98	1.35	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	4.76	4.23
2	1995-01-01 00:02:00	0.24	0.99	-1.02	1.45	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.75	4.26
3	1995-01-01 00:03:00	0.19	1.02	-0.98	1.44	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.75	4.29
4	1995-01-01 00:04:00	-0.14	0.91	-0.81	1.25	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.74	4.33
5	1995-01-01 00:05:00	0.09	0.94	-0.78	1.26	-310.30	-1.60	25.50	311.40	18.46	17347.00	3.58	0.16	3.74	4.36
6	1995-01-01 00:06:00	9999.99	9999.99	9999.99	9999.99	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.74	5.28
7	1995-01-01 00:07:00	9999.99	9999.99	9999.99	9999.99	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.73	3.86
8	1995-01-01 00:08:00	9999.99	9999.99	9999.99	9999.99	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.73	3.89
9	1995-01-01 00:09:00	9999.99	9999.99	9999.99	9999.99	99999.90	99999.90	99999.90	99999.90	999.99	9999999.00	99.99	999.99	3.72	3.92

6





1. The datasets should be downloaded again
2. Ignore this and continue
3. This is a big deal, and finding a solution is essential

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1. Vectorization with NumPy

Pure Python:

```
python

result = []
for i in range(len(a)):
    result.append(a[i] + b[i])
```

Using NumPy:

```
python

import numpy as np
a = np.array(a)
b = np.array(b)
result = a + b
```

2. Data Manipulation with Pandas

Pure Python:

```
python

result = {}
for key in data:
    result[key] = sum(data[key])
```

Using pandas:

```
python

import pandas as pd
df = pd.DataFrame(data)
result = df.sum()
```

3. Scientific Computing with SciPy:

- SciPy is built on NumPy.
- SciPy provides more utilities for:
 - optimization, linear algebra, integration, interpolation, special functions, FFT, signal and image processing, ODE solvers, and more.

4. Parallel Computing with joblib or Dask:

- Speed up processing times.

5. Just-In-Time Compilation with Numba:

- Best suited for **mathematical functions**, like those used in machine learning and data analysis.

6. Efficient Array Computing with JAX:

- Powerful transformations.

7. Profiling:

- Allows you to focus your **optimization** efforts on the **slowest parts** of your code.

The authors:

- Focus on ACE satellite data. (1998 -2020)
- Used AI techniques such as machine learning and statistical inference to analyze the ACE data and gain insights into the solar wind.
- They found that some parameter distributions are well approximated by Gaussian distributions while others are closer to lognormal laws.

Important: Pre-processing

- The magnitude of Values
- **Spread values** seem **dangerous** to implement in a **deep-learning** algorithm **without a pre-process**.

Towards an AI-based understanding of the solar wind: A critical data analysis of ACE data

S. Bouriat^{1,2,3,4*}, P. Vandame³, M. Barthélémy^{1,2} and J. Chanussot³

¹CNRS, IPAG, University of Grenoble Alpes, Grenoble, France, ²CSUG, University of Grenoble Alpes, Grenoble, France, ³GIPSA-Lab, Grenoble INP, CNRS, University of Grenoble Alpes, Grenoble, France, ⁴SpaceAble, Paris, France

1. Feature Scaling:

- It fits within a specific scale, like 0-100 or 0-1.

2. Understanding Data:

- gaining insights from the statistical properties of the data

3. Outlier Detection:

- They might be due to variability in the data.
- They may indicate experimental errors.

4. Normalization:

- It is a specific type of feature scaling
- Normalization typically means rescaling the values into a range of [0,1].

5.1. Description

Description

Variable	Description	Note	Magnitude
date_time	year, day, hour, minute		
By			-10 to 10 (nT)
Bx			-10 to 10 (nT)
Bz			-10 to 10 (nT)
BT	Field magnitude average		1 to 20 (nT)
Vx			250 to 1000 (km/s)
Vy			250 to 1000 (km/s)
Vz			250 to 1000 (km/s)
VT	Speed		250 to 1000 (km/s)
n_p	Proton density		1 to 10 (p/cc)
T_p	Proton temperature		10000 to 2*10^6 (K)
P_p	Flow pressure		1 to 10 (nPa)
E	Electric field		0 to 10 (mV/m)
dBz	dbz_nez		-100 to 100 (nT)
dBe	dbe_nez	dropped later	-100 to 100 (nT)
dBn	dbn_nez	dropped later	-100 to 100 (nT)
dBH	$\sqrt{(dBe)^2 + (dBn)^2}$	Engineered	-100 to 100 (nT)

Note:

- There is **NO** relationship between **Bz** and **dBz**, just similar names.

5. Improving Accuracy:

- Scaling features to a similar range prevents any single feature from dominating the model simply because of its scale.

6. To understand the physical processes involved:

- The magnitudes of the variables aren't just numbers - they represent physical quantities tied to real-world processes.

5.1. Description

Description

Variable	Description	Note	Magnitude
date_time	year, day, hour, minute		
By			-10 to 10 (nT)
Bx			-10 to 10 (nT)
Bz			-10 to 10 (nT)
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Vx			250 to 1000 (km/s)
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Vz			250 to 1000 (km/s)
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n_p	Proton density		1 to 10 (p/cc)
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E	Electric field		0 to 10 (mV/m)
dBz	dbz_nez		-100 to 100 (nT)
dBe	dbe_nez	dropped later	-100 to 100 (nT)
dBn	dbn_nez	dropped later	-100 to 100 (nT)
dBH	$\sqrt{(dBe)^2 + (dBn)^2}$	Engineered	-100 to 100 (nT)

Note:

- The is **NO** relationship between **Bz** and **dBz**, just similar names.

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- **Data Analysis and Code Development**

- **Analyzing Missing Data Patterns:** I dedicated time to work on code related to understanding the distribution of missing values across different years in our dataset. This analysis will help us gain valuable insights into potential data gaps and allow us to make informed decisions for data preprocessing.

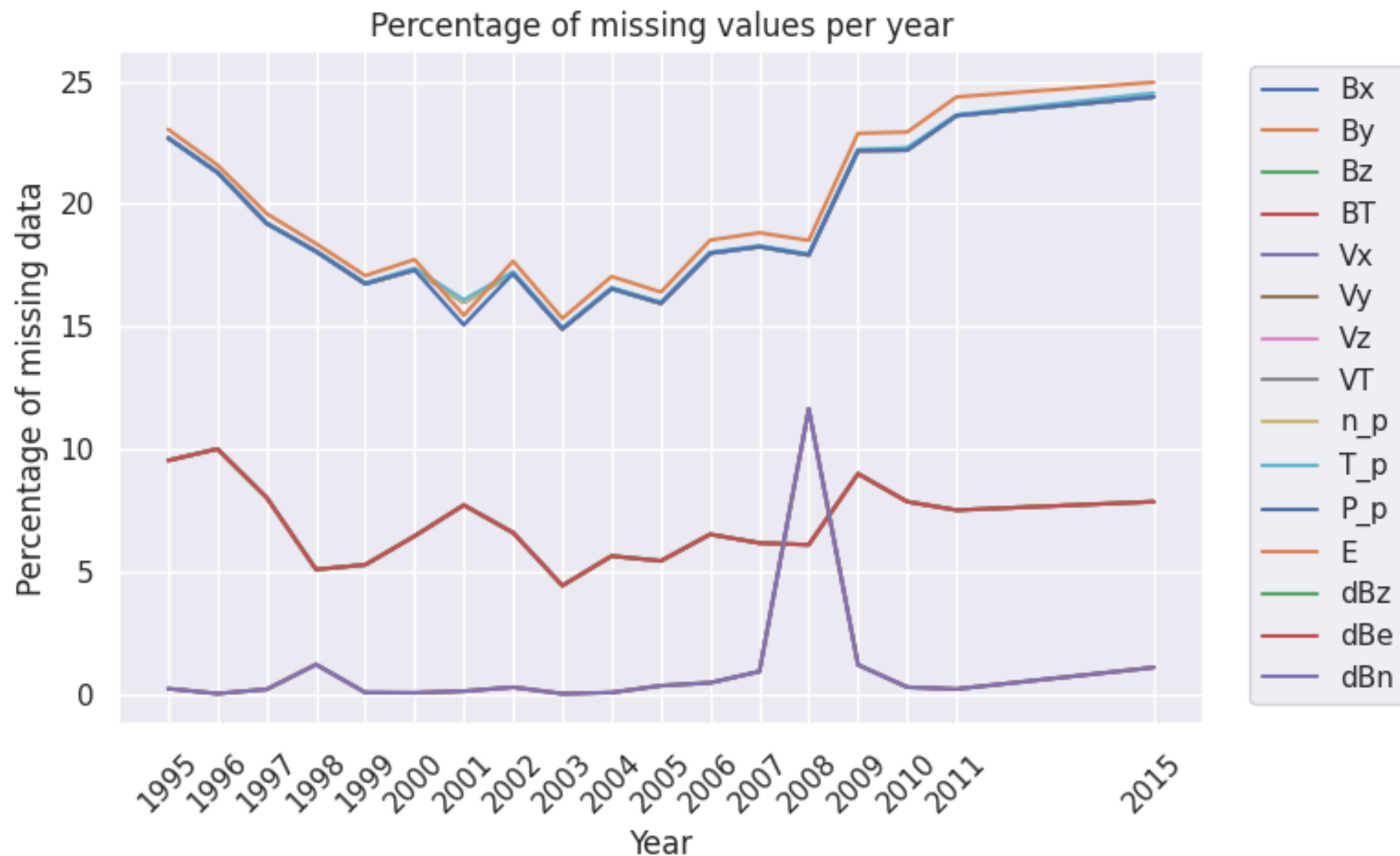
- **Literature Review**

- **Researching Previous Works on Missing Data:** I spent significant time searching and reading research papers that focus on handling missing data. By studying previous works, I aimed to learn from existing methodologies and best practices to improve our approach in dealing with missing data effectively.
- **Gap Identification in Space Science Research:** While there were numerous papers discussing and comparing missing data handling methods in various fields, including healthcare, finance, and social sciences, I could not find any relevant studies specific to the field of space science. Identifying this gap in the literature is crucial as it presents an opportunity for us to contribute to the domain with our research and potentially adapt and evaluate existing methodologies from other fields to address the challenges unique to space science data.

- **Hyperparameter Optimization Study**

I am studying hyperparameter optimization techniques. I haven't finished it yet. These methods play a crucial role in fine-tuning machine learning models, and by enhancing our understanding of hyperparameter optimization, we can enhance the overall performance of our models.

2



- Singular Spectrum Analysis (SSA) is a method for filling in large gaps in the solar wind and IMF data.
- SSA works by reconstructing missing data using an iteratively inferred, smooth "signal" that captures coherent modes, while "noise" is discarded.
- SSA can infer smooth temporal modes of co-variability between the solar driver and the response from the existing data.
- By combining SSA with geomagnetic indices that are time-continuous, we can fill in large gaps in the solar wind and IMF data.
- SSA is a powerful tool for filling in gaps in data, but it is not perfect. The reconstructed data may not be as accurate as the original data, and there may be some errors.
- SSA can be used to fill in gaps in a variety of data types, not just solar wind and IMF data.

Gap filling of solar wind data by singular spectrum analysis

D. Kondrashov,¹ Y. Shprits,¹ and M. Ghil^{1,2}

Received 28 May 2010; accepted 17 June 2010; published 4 August 2010.

[1] Observational data sets in space physics often contain instrumental and sampling errors, as well as large gaps. This is both an obstacle and an incentive for research, since continuous data sets are typically needed for model formulation and validation. For example, the latest global empirical models of Earth's magnetic field are crucial for many space weather applications, and require time-continuous solar wind and interplanetary magnetic field (IMF) data; both of these data sets have large gaps before 1994. Singular spectrum analysis (SSA) reconstructs missing data by using an iteratively inferred, smooth "signal" that captures coherent modes, while "noise" is discarded. In this study, we apply SSA to fill in large gaps in solar wind and IMF data, by combining it with geomagnetic indices that are time-continuous, and generalizing it to multivariate geophysical data consisting of gappy "driver" and continuous "response" records. The reconstruction error estimates provide information on the physics of co-variability between particular solar-wind parameters and geomagnetic indices. **Citation:** Kondrashov, D., Y. Shprits, and M. Ghil (2010), Gap filling of solar wind data by singular spectrum analysis, *Geophys. Res. Lett.*, 37, L15101, doi:10.1029/2010GL044138.

Broadly speaking, these indices can be considered as a proxy for the overall time-lagged *magnetospheric response* to the *solar driver*, i.e., to the solar wind and IMF.

[5] The purpose of this paper is to reconstruct data in the gaps of the solar driver by using smooth spatio-temporal modes of co-variability inferred by singular spectrum analysis (SSA) from time-lagged correlations in multivariate data—the data sets consisting of various geomagnetic indices, solar wind and IMF—while discarding the noise. Kondrashov and Ghil [2006] developed an SSA-based gap-filling method with applications that rely, so far, mainly on the presence of significant oscillatory modes in the time series [see Kondrashov *et al.*, 2005]. We show here that this method can also be successfully applied to multivariate geophysical data sets that consist, broadly speaking, of gappy-driver and continuous-response records, by relying on analogous episodes of co-variability which is not necessarily periodic.

[6] Regression based techniques [Vassiliadis *et al.*, 1995; Chen and Sharma, 2006] can be used to derive predictive filter that relate time series of lagged input from the past, i.e., solar wind parameters, to the current values of magneto-

This research paper proposes a novel splitting-based **IMV-RE (Imputing the Missing Values in Real-Time Environment) algorithm to impute different missing values within a dataset.**



Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

A Novel Algorithm for Imputing the Missing Values in Incomplete Datasets

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Sant Longowal Institute of Engineering and Technology

Manminder Singh (✉ manminderfzr@yahoo.com)

Sant Longowal Institute of Engineering and Technology

How does the IMV-RE algorithm work, and what makes it different from other imputation methods?

- The IMV-RE algorithm is a novel splitting-based data imputation algorithm that can predict missing values efficiently and accurately.
- The algorithm first decides the upper limit for each feature within a cluster, which is further used for imputing the missing values so that the imputed values lie much nearer to the exact values.
- Unlike other imputation techniques, the IMV-RE algorithm only searches within the class to calculate the centered value for the final imputation, which makes it more efficient for real-time problems.
- The proposed IMV-RE algorithm successfully abstracts the dependency of the CCMVI technique on the complete samples within a class to estimate the missing values.
- The experimentation on ten benchmark datasets indicates that the proposed IMV-RE algorithm outperforms the existing techniques in sensitivity to Accuracy, Root Mean Square Error (RMSE), and Coefficient of Determination (R²).
- The IMV-RE algorithm is a robust statistical MVI technique that can handle missing values in a variety of data types.
- The algorithm is efficient and can be used for real-time problems.
- The IMV-RE algorithm has been shown to outperform other imputation techniques in terms of accuracy, RMSE, and R².

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The authors dissect the pros and cons of various methods to handle these missing values, ultimately recommending estimation techniques as the optimal solution for time series data.

✓ Conventional Methods: (Mean and Mode imputation, and deletion)

- Advantages:
 - Simplicity:** They are easy to understand and implement.
 - Generalizability:** Can be applied to a variety of datasets, not just time series data.
 - Effectiveness for Low Missingness:** These methods can yield satisfactory results when the percentage of missing data is low.
- Disadvantages:
 - Limited Effectiveness:** Their effectiveness wanes when there's a high percentage of missing data.
 - Bias:** They can produce biased results, especially if the missingness is systematic or related to any inherent property of the data.
 - Loss of Data:** Methods like deletion result in a loss of data, leading to reduced efficiency.
 - Skewed Estimation:** The resultant dataset after handling missing values can lead to misleading and biased analyses.

✓ Estimation Methods:

- Autoregressive-model-based missing values estimation method
- Optimized fuzzy c-means with support vector regression and genetic algorithm
- Fuzzy-rough methods
- Regression analysis
- Support vector regression
- Genetic algorithms

A Review of Missing Values Handling Methods on Time-Series Data

Irfan Pratama¹, Adhistya Erna Permanasari², Igi Ardiyanto³, Rini Indrayani⁴
Department of Electrical Engineering and Information Technology
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Abstract — Missing values becomes one of the problems that frequently occur in the data observation or data recording process. The needs of data completeness of the observation data for the uses of advanced analysis becomes important to be solved. Conventional method such as mean and mode imputation, deletion, and other methods are not good enough to handle missing values as those method can caused bias to the data. Estimation or imputation to the missing data with the values produced by some procedures or algorithms can be the best possible solution to minimized the bias effect of the conventional method of the data. So that at last, the data will be completed and ready to use for another step of analysis or data mining. In this paper, we will explain and describe several previous studies about missing values handling methods or approach on time series data. This paper also discuss some plausible option of methods to estimate missing values to be used by other researchers in this field of study. The discussion's aim is to help them to figure out what method is commonly used now along with its advantages and drawbacks.

because of an observation devices breakdown. While in an opinion survey or an interview, missing values caused by respondents who decline to answer or complete the survey, or even because insufficient survey observation's stuff [2,3]. Both examples stated above are trully unexpected occurrence. In other word, the data are expected to be observed (sensor failure cases), yet it is missing due to the setbacks. Missing values is probably can not be evaded as it is happened unexpectedly. Time series data are become one of some data that most likely to have missing values in it.

Time series is a data that observed in regular interval of times [4]. Time series form can found as a continous data or discrete because observed in a regular interval of time (example: daily, monthly, annualy). Some examples of continous and discrete time series data are sinusoidal signal which is continous time series data, while daily stock prices, temperature data are a discrete data.

Data measurements are conducted several times with different condition, and sometimes, missing data occurred due to several problem that are known as the "missingness mechanism" [5,6,7]:

1) Missing completely at random (MCAR): a variable

- Advantages:
 - **Bias Reduction:** They are designed to minimize biases introduced by conventional methods.
 - **Accuracy:** Can provide more precise estimates for missing values.
 - **Flexibility:** Methods such as genetic algorithms and fuzzy c-means can be tailored according to user needs.
 - **Complementary Use:** Can be combined with other methods to enhance accuracy.
 - **Handling Large Missing Data:** They are especially suited for datasets with a significant amount of missing values or non-ignorable missing data.
 - **Specialization for Time Series:** Many estimation techniques are specifically designed to consider the temporal relationships in time series data.
- Disadvantages:
 - **Potential for Bias:** While they reduce bias compared to conventional methods, they aren't entirely free from introducing their own biases.
 - **Computational Intensity:** Some methods can be computationally intensive, requiring more time and computational resources.
 - **Variable Performance:** The effectiveness of a method can vary depending on the specific characteristics of the dataset.
 - **Complex Implementation:** Some techniques require a deep understanding of the underlying algorithms, making them challenging for the average user.

✓ Conclusion:

- Handling missing values is a critical step in data preprocessing, especially in time series data.
- Conventional methods, can be simple but may not be ideal for datasets with a large number of missing values or non-random missingness.
- Estimation techniques, offer a more robust and flexible approach to dealing with missing data in such scenarios.

A Review of Missing Values Handling Methods on Time-Series Data

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Data measurements are conducted several times with different condition, and sometimes, missing data occurred due to several problem that are known as the "missingness mechanism" [5,6,7]:

1) Missing completely at random (MCAR): a variable

The authors compare five methods evaluated are:

- listwise deletion
- mean substitution
- regression imputation
- stochastic regression imputation
- multiple imputations

Conclusion:

- Multiple imputation is an effective tool for addressing missing data in logistic regression analyses.
- Multiple imputation can be used to handle both MCAR and MAR data types.
- Listwise deletion is not biased for MCAR data, but it can be biased for MAR data.
- Multiple imputation is generally preferred over listwise deletion for MAR data.
- Multiple imputation is especially recommended when dealing with up to 20% missing data.

Logistic Regression with Missing Data: A Comparison of Handling Methods, and Effects of Percent Missing Values

Sutthipong Meeyai

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Email: sutthi@sut.ac.th

Abstract—The aim of this article is to compare five popular missing data handling methods: listwise deletion, mean substitution, regression imputation, stochastic imputation, and multiple imputation. Three missing data mechanisms are investigated: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). A Monte Carlo simulation is applied to simulate data and then logistic regression parameters are estimated. Our findings show that, among the five missing data handling methods, multiple imputation performs well on both MCAR and MAR. There is no evidence indicating that listwise deletion and multiple imputation produce biased parameters for MCAR. None of these techniques can handle MNAR. Finally, this article suggests maximum percent missing data and a sample size for listwise deletion and multiple imputation techniques.

about the amount of missing data or showing how missing data were handled in their studies. Davey and Savla [8] also noted that the implications of missing data for social research had not received well-known treatment to date.

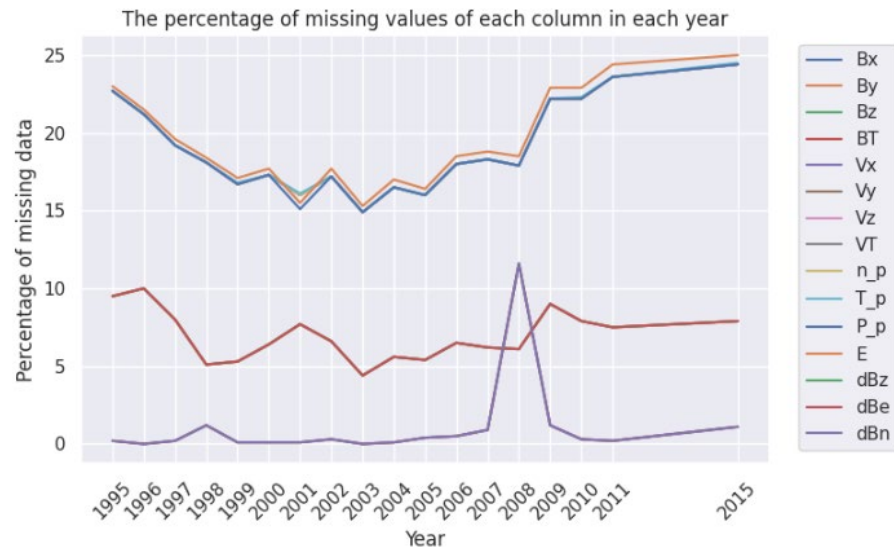
Although the importance of missing data have long been realized, the comparison of missing data handling methods for a logistic regression is rarely discussed. The aim of this research is to provide researchers guidelines to choose appropriate missing data handling techniques for logistic regression analysis and to suggest maximum percent missing data and a sample size.

II. THEORETICAL BACKGROUND

- **Enhanced Code Readability and Transparency:**
 - Adding informative comments and explanations.
 - Advantages:
 - **Improved maintenance:** Clean and well-documented code is easier to maintain.
 - **Facilitated collaboration:** Readable code ensures that colleagues can understand, modify, and work on the code without constantly consulting the original author.
 - **Reduced errors:** Clear and transparent code reduces the likelihood of introducing errors.
 - **Increased code longevity:** Readable code has a longer lifespan.
 - **Better onboarding:** New team members can get up to speed faster if the code is transparent and well-commented.
 - **Reduced dependency:** The project becomes less dependent on the original coder.
 - **Professionalism:** Well-documented code reflects professionalism.

Two types of years:

- leap years: 366 days
- non-leap years: 365 days



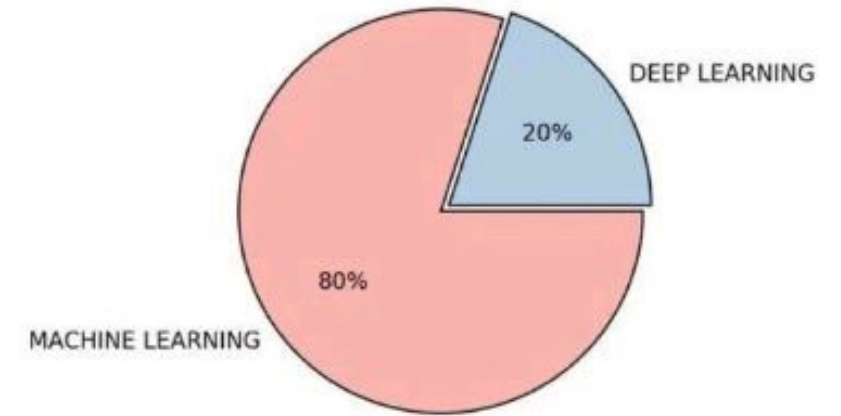
```
GIC Data Exploration.ipynb
File Edit View Insert Runtime Tools Help Last saved at 12:06 AM

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1. Libraries & paths & functions
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2.2. SuperMAG data
3. Data acquisition and compression
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3.1.1. Description
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3.2. SuperMAG data
3.2.1. All years
3.2.1.1. Description
3.2.1.2. Code
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4. Merge datasets
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5.1. Description
5.2. Data types

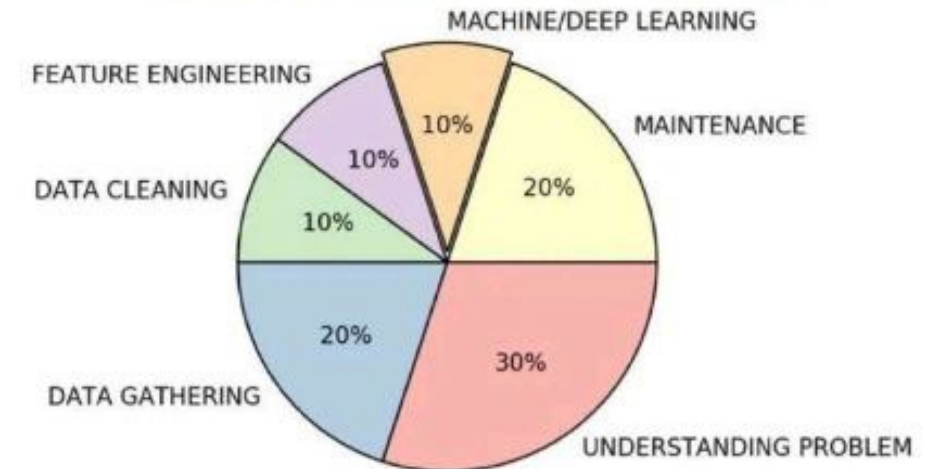
217 return df_missing_values_report
218
219
220 # 5. minutes_in_years
221 def minutes_in_years(start, end):
222     """
223     min per year:
224     There are two types of years:
225     leap years: 366 days
226     non-leap years: 365 days
227
228     The rules for determining whether a year is a leap year or not are as follows:
229     1. If a year is evenly divisible by 4, it is a leap year, except:
230     2. If that year is also evenly divisible by 100, it is not a leap year, unless:
231     3. That year is also evenly divisible by 400, in which case it is a leap year.
232     So while the year 100 is evenly divisible by 4, it is also evenly divisible by 100.
233     Since it is not evenly divisible by 400, it does not meet the third criteria, and therefore it is not a leap year.
234
235     These rules are in place to correct for the fact that a solar year (the length of time it takes for the Earth to orbit the Sun once) is not exactly 365.25 days,
236     but is approximately 365.2425 days.
237     By skipping the leap day in years divisible by 100 (but not by 400), the Gregorian calendar maintains its alignment with the solar year over the centuries.
238
239     0.0025 (day) * 400 = 1 (day) So, days = 366 if ((year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)) else 365
240
241     # use the function
242     >>> for year, minutes in minutes_in_years(1995, 2015):
243     >>>     print(f"{year}: {minutes}")
244
245     1995: 525600
246     1996: 527040
247     1997: 525600
248     1998: 525600
249     1999: 525600
250     2000: 527040
251     2001: 525600
252     2002: 525600
253     2003: 525600
254     2004: 527040
255     2005: 525600
256     2006: 525600
257     2007: 525600
258     2008: 527040
259     2009: 525600
260     2010: 525600
261     2011: 525600
262     2012: 527040
263     2013: 525600
264     2014: 525600
265     2015: 525600
266
267     minutes_per_day = 24 * 60
268     for year in range(start, end + 1):
269         days = 366 if ((year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)) else 365
270         yield year, days * minutes_per_day
271
```


- **Misconception:**
 - Machine Learning 80%
 - Deep Learning 20%
- **In reality:**
 - Pre-processing 90%
 - Machine and Deep Learning 10%
- **My work:**
 - Completed the pre-processing phase, which is the bulk of the work.
 - My goal: focusing on learning about Machine and Deep learning models. Then commence with the modeling.
- **Models:**
 - Hyperparameter Optimization
 - LSTM
 - CNN
 - SVM

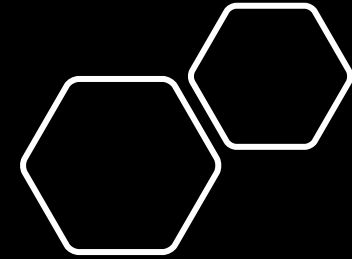
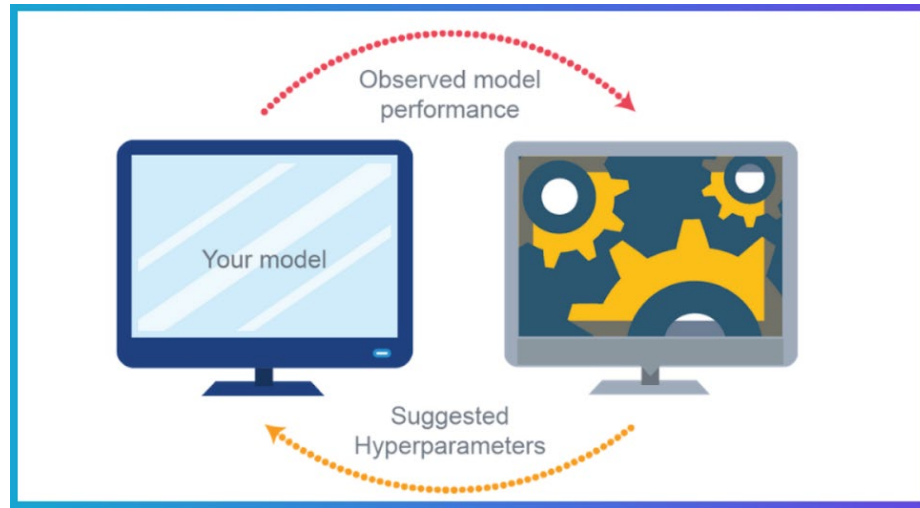
DATA SCIENTIST JOB - EXPECTATION



DATA SCIENTIST JOB - REALITY



Hyperparameter Optimization methods

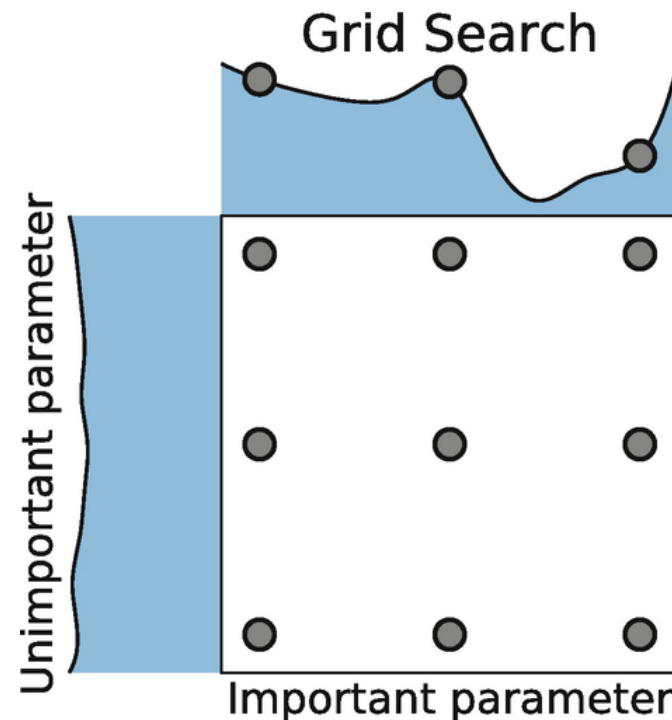


- **Concepts**

- Hyperparameters
 - Batch size, Epochs, Dropout rate, Learning Rate, Optimizers
- Cross join

- **Optimization methods**

- 1. Gradient-based optimization
- 2. Grid search
- 3. Random search
- 4. Bayesian optimization
- 5. Adaptive search
- 6. Early stopping



In Machine Learning:

- **Parameters** are estimated from data **automatically**. (dependent variable)

Examples:

- Beta(s) in linear/logistic regression
- support vectors in SVM
- weights in NN

- **Hyperparameters** are set **manually** and are used in processes to help estimate model parameters. (independent variable)

Examples:

- k in KNN
- C and sigma in SVM
- learning rate in NN

Do not confuse this nominal similarity:

In Statistics: Like the Gaussian distribution parameters (mean (μ) and standard deviation (σ)).

In Programming: You may pass a parameter to a function.

Hyperparameters in neural networks

3

Batch_size	Epochs	Embedding_dim	Hidden_size	Dropout_rate	Learning_Rate	Optimizer
32	2	16	32	0.1	0.0001	SGD
64	3	32	64	0.2	0.001	Momentum
128	4	64	128	0.3	0.01	Nesterov Accelerated Gradient (NAG)
256	10	128	256	0.4	0.1	
				0.5		Adagrad
				0.6		Adadelata
				0.7		RMSprop
				0.8		Adam
				0.9		AdaMax
						Nadam

Batch size: The number of training examples utilized in one [iteration](#) (The number of batches = The number of times the algorithm's parameters are updated).

Epochs: Indicates the number of passes of the entire training dataset the machine learning algorithm has completed.

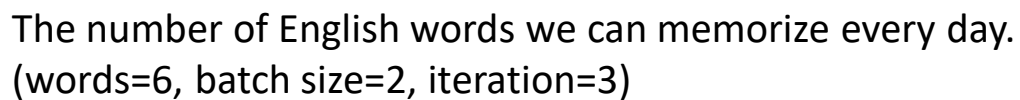
Embedding dim: The number of columns of the embedding layer which represents words as numbers.

Hidden size: The number of hidden neurons which it should be between the size of the input layer and the size of the output layer.

Dropout rate: A regularization technique for reducing overfitting in artificial neural networks by turning off a portion of the nodes (In other words not to update a portion of weights).

Learning rate: Determines the step size at each iteration while moving toward a minimum of a loss function.

Optimizer: An algorithm to optimize gradient descent.



- Overfitting: Learning one kind of tree well but not experienced on the other trees.

Parameters		Combinations	
10	1	10	1
20	2	10	2
30		20	1
		20	2
		30	1
		30	2

$$3 * 2 = 6$$

EXPENSIVE

$$4 * 4 * 4 * 4 * 9 * 4 * 9 = 82,944$$

Batch_size	Epochs	Embedding_dim	Hidden_size	Dropout_rate	Learning_Rate	Optimizer
32	2	16	32	0.1	0.0001	SGD
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				0.6		RMSprop
				0.7		Adam
				0.8		AdaMax
				0.9		Nadam

1. Gradient-based optimization
2. Grid search
3. Random search
4. Bayesian optimization
5. Adaptive search
6. Early stopping

“The machine learning engineer is squarely in the hyperparameter tuning phase of the model development lifecycle, asking themselves: would different features, model architectures, or learning process hyperparameters produce a better model?”

Ref:

Hyperparameter optimization https://en.wikipedia.org/wiki/Hyperparameter_optimization

Adaptive Hyperparameter Tuning <https://determined.ai/assets/images/resources/AdaptiveHyperparameterTuning.pdf>

An overview of gradient descent optimization algorithms

2 Gradient descent variants

There are three variants of gradient descent, which differ in how much data we use to compute the gradient of the objective function. Depending on the amount of data, we make a trade-off between the accuracy of the parameter update and the time it takes to perform an update.

2.1 Batch gradient descent

2.2 Stochastic gradient descent

2.3 Mini-batch gradient descent

Mini-batch gradient descent is typically the algorithm of choice when training a neural network and the term **SGD** usually is employed also when mini-batches are used.

4 Gradient descent optimization algorithms

4.1 Momentum

4.2 Nesterov Accelerated Gradient

4.3 Adagrad

4.4 Adadelta

4.5 RMSprop

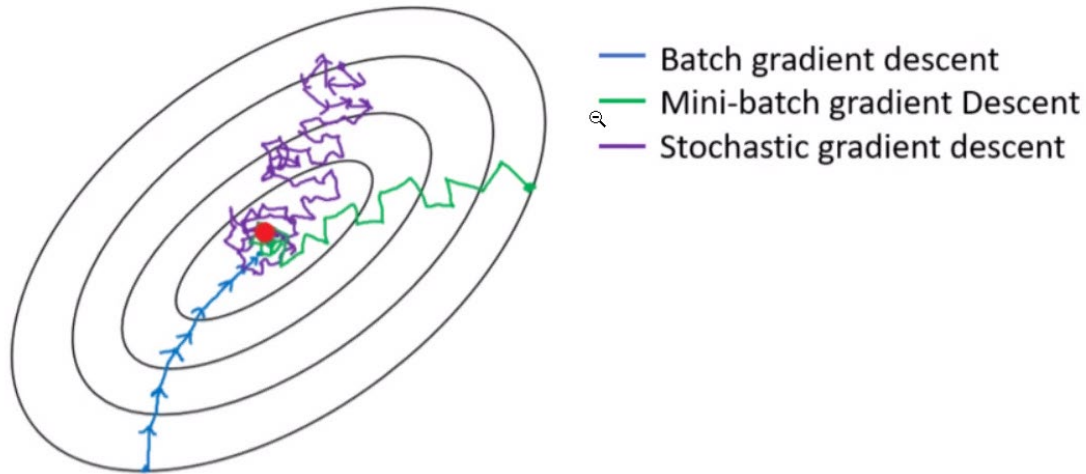
4.6 Adam

4.7 AdaMax

4.8 Nadam

Ref: 1. <https://arxiv.org/abs/1609.04747>

2. <https://www.analyticsvidhya.com/blog/2021/06/complete-guide-to-gradient-based-optimizers/>



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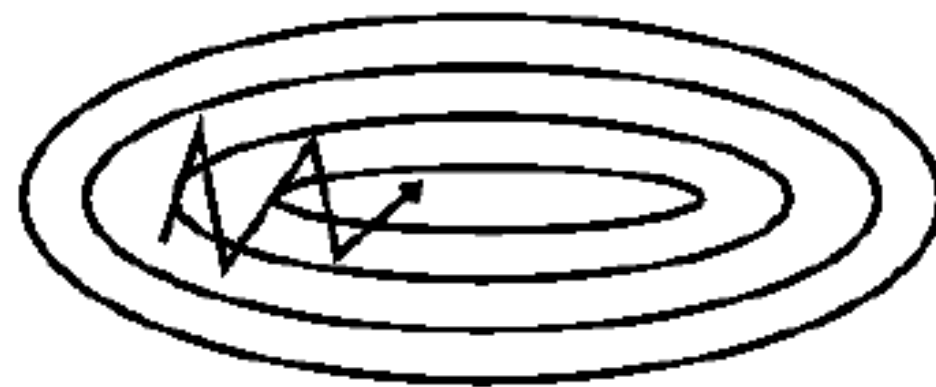
4.6 Adam

4.7 AdaMax

4.8 Nadam



(a) SGD without momentum



(b) SGD with momentum

Figure 2: Source: [Genevieve B. Orr](#)

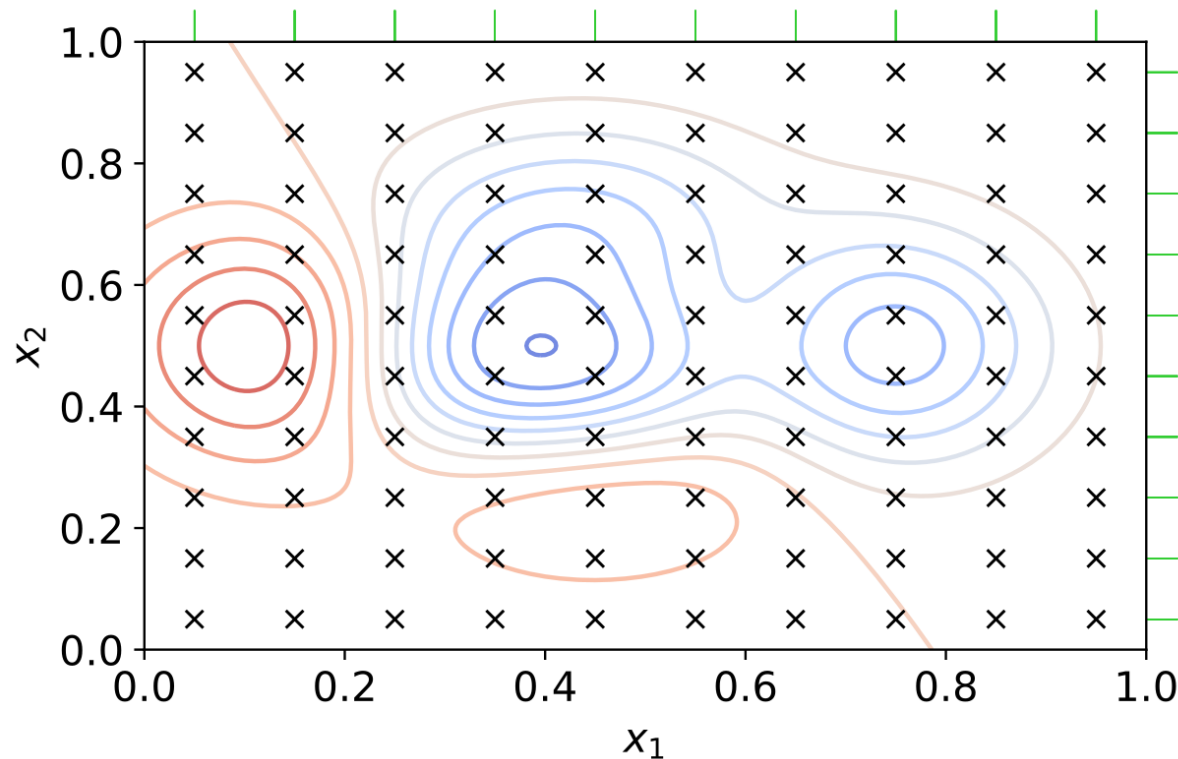
Essentially, when using momentum, we push a ball down a hill. The ball accumulates momentum as it rolls downhill, becoming faster and faster on the way (until it reaches its terminal velocity, if there is air resistance, i.e. $\gamma < 1$). The same thing happens to our parameter updates: The momentum term increases for dimensions whose gradients point in the same directions and reduces updates for dimensions whose gradients change directions. As a result, we gain faster convergence and reduced oscillation.

2. Grid search

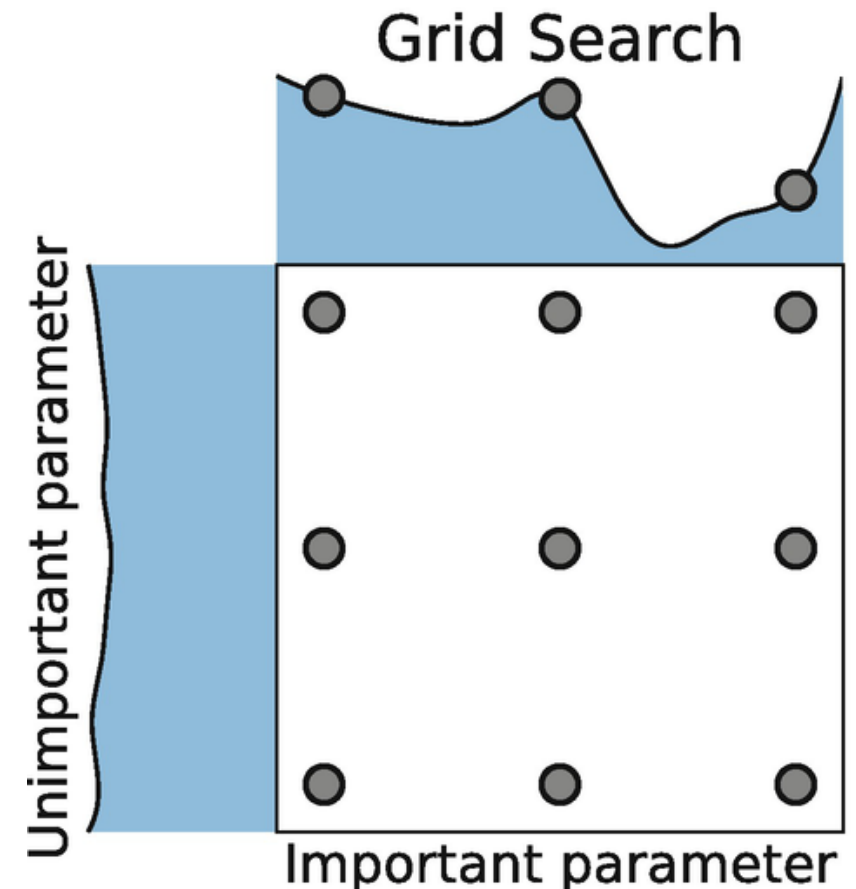
10

Grid search across different values of two hyperparameters. For each hyperparameter, **10** different values are considered, so a total of **100** different combinations are evaluated and compared.

- Blue indicates regions with strong results.
- Red contours show regions with poor results.



(1, 1)	(1, 2)	(1, 3)
(2, 1)	(2, 2)	(2, 3)

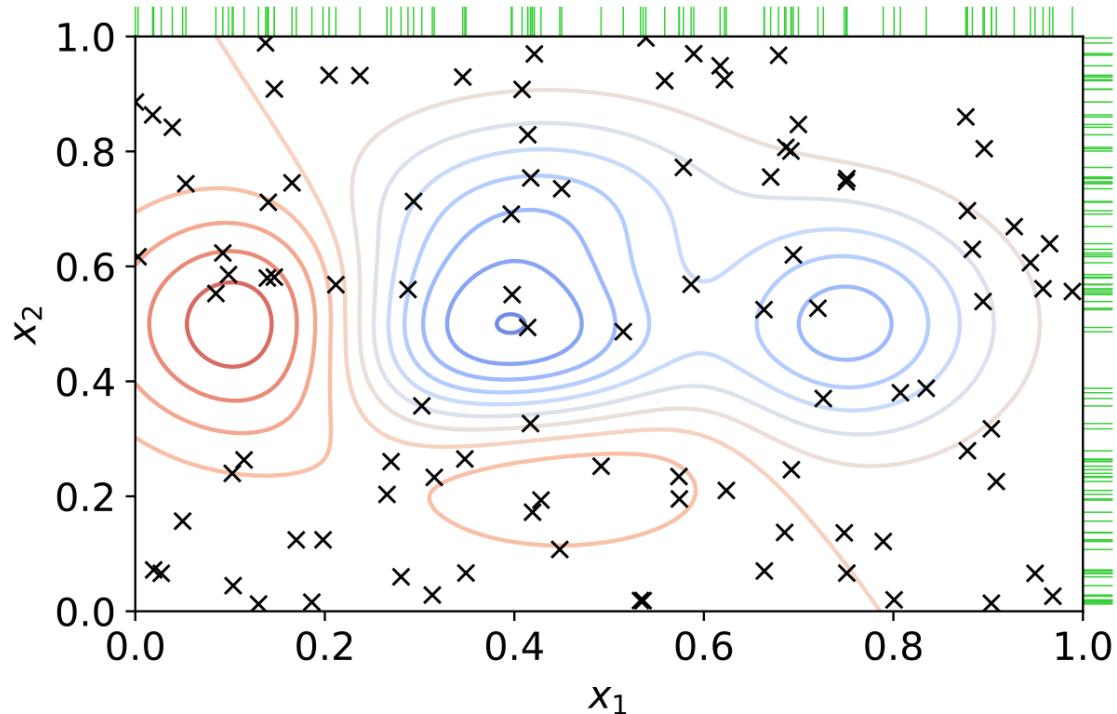


3. Random search

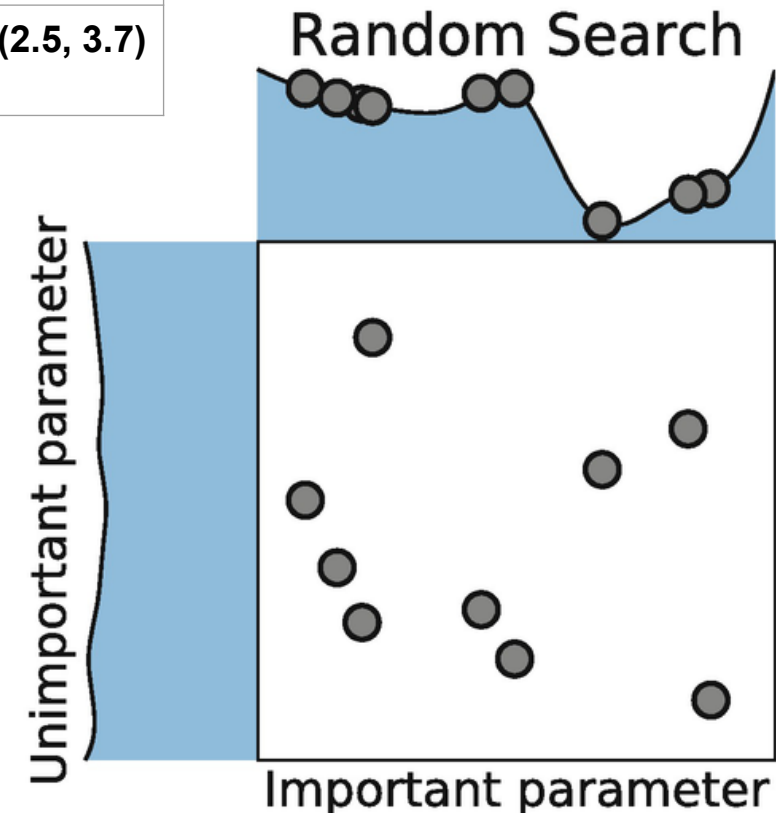
11

Random search across different combinations of values for two hyperparameters. In this example, 100 different random choices are evaluated. The **green bars** show that **more individual values for each hyperparameter are considered** compared to a grid search.

- Blue indicates regions with strong results.
- Red contours show regions with poor results.



(1, 1)	(1.2, 1.8)	(0.8, 3.3)
(2.3, 0.9)	(2.1, 2.2)	(2.5, 3.7)



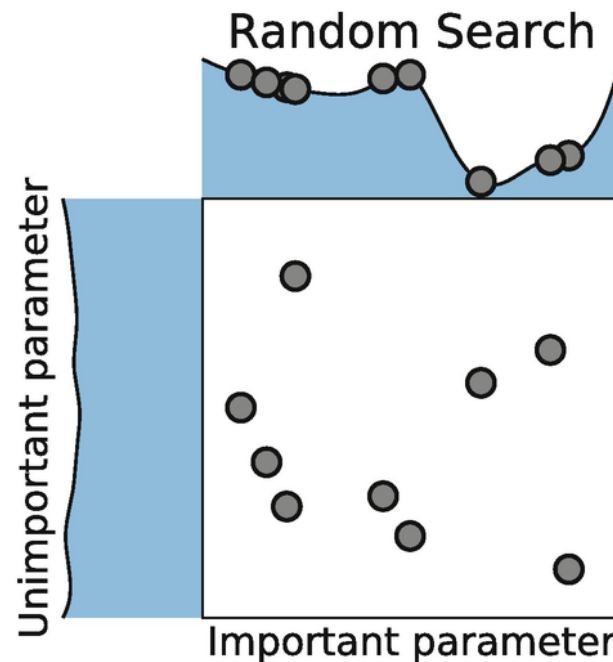
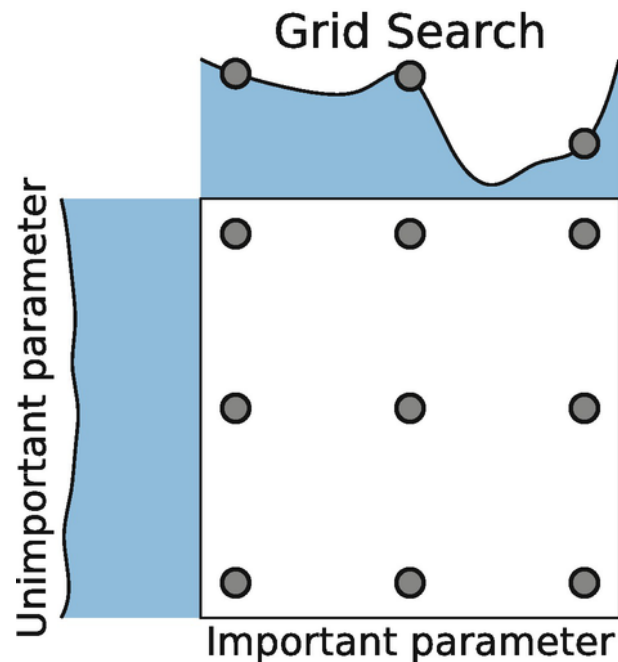
Random search outperforms Grid search

12

Random search outperforms Grid search because in Random search more individual values for each hyperparameter are considered compared to a grid search.

(1, 1)	(1, 2)	(1, 3)
(2, 1)	(2, 2)	(2, 3)

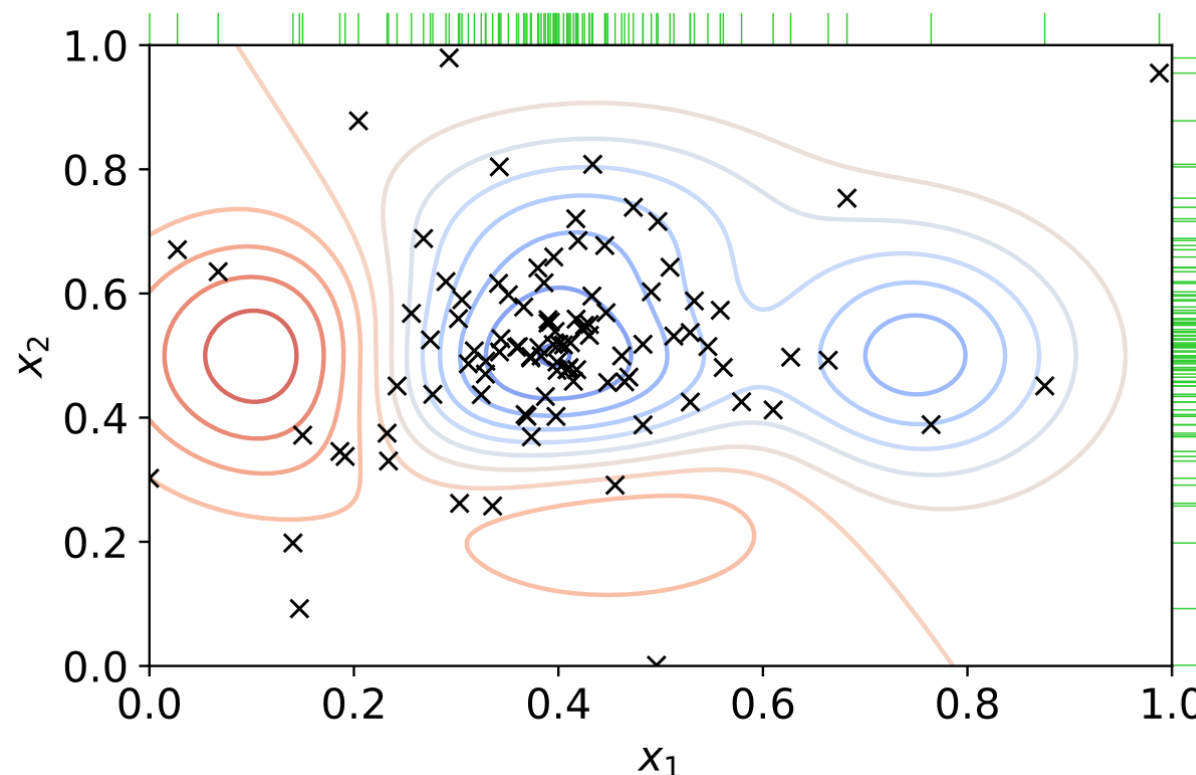
(1, 1)	(1.2, 1.8)	(0.8, 3.3)
(2.3, 0.9)	(2.1, 2.2)	(2.5, 3.7)



Methods such as Bayesian optimization **smartly explore the space** of potential choices of hyperparameters **by deciding which combination to explore next based on previous observations**.

In practice, Bayesian optimization has been shown to obtain **better results in fewer evaluations compared to grid search and random search**, due to the ability to reason about the quality of experiments before they are run.

- Blue indicates regions with strong results.
- Red contours show regions with poor results.
- The green bars show that **more individual values for each hyperparameter are considered** compared to a grid search.



- Adaptive search is not a general approach.
- Main idea: We are not willing to watch a game that one side is winning by a large margin.

Although there are a few works that adaptively optimize the hyperparameters of specific algorithms depending on the environment (e.g., adaptive octree-resolution selection for visual SLAM [17]), to the best of our knowledge, no work has proposed a general and adaptive hyperparameter tuning framework without a limitation on the algorithms to be tuned.

Adaptive Hyperparameter Search

We all know what it's like to watch a game on TV when one side is winning by such a large margin that the game is "over before it's over." While a comeback is possible, it just doesn't feel worth watching for another half hour to the end, so we grab the remote and flip to another game. Just as we might

change the channel in this scenario, Determined's adaptive hyperparameter search abandons unpromising hyperparameter configurations that are unlikely to outperform other configurations under consideration.

Ref: Adaptive Hyperparameter Tuning

1. <https://arxiv.org/abs/2107.00275>
2. <https://determined.ai/assets/images/resources/AdaptiveHyperparameterTuning.pdf>

There are seven hyperparameters here:

Step 1. We use the value of the first row for the first six hyperparameters but all values of the seventh hyperparameter (optimizer).

Step 2. The calculations continue with the first row of the first five hyperparameters, the second row of the sixth hyperparameter (learning rate), and all rows of seven hyperparameters (optimizer).

Step 3. We repeat the second step, except that we choose the third row and again the fourth row for the sixth parameter (learning rate).

Step 4. The results show that "Adam" had much better results than other optimizers when calculating all 36 combinations. So we only use Adam as the optimizer for doing other scenarios.

Batch_size	Epochs	Embedding_dim	Hidden_size	Dropout_rate	Learning_Rate	Optimizer	Accuracy	Time	Loss
32	2	16	32	0.1	0.0001	SGD			
64	3	32	64	0.2	0.001	Momentum			
128	4	64	128	0.3	0.01	Nesterov Accelerated Gradient (NAG)			
256	10	128	256	0.4	0.1	Adagrad			
				0.5		Adadelta			
				0.6		RMSprop			
				0.7		Adam			
				0.8		AdaMax			
				0.9		Nadam			

A major challenge in NN: Training time

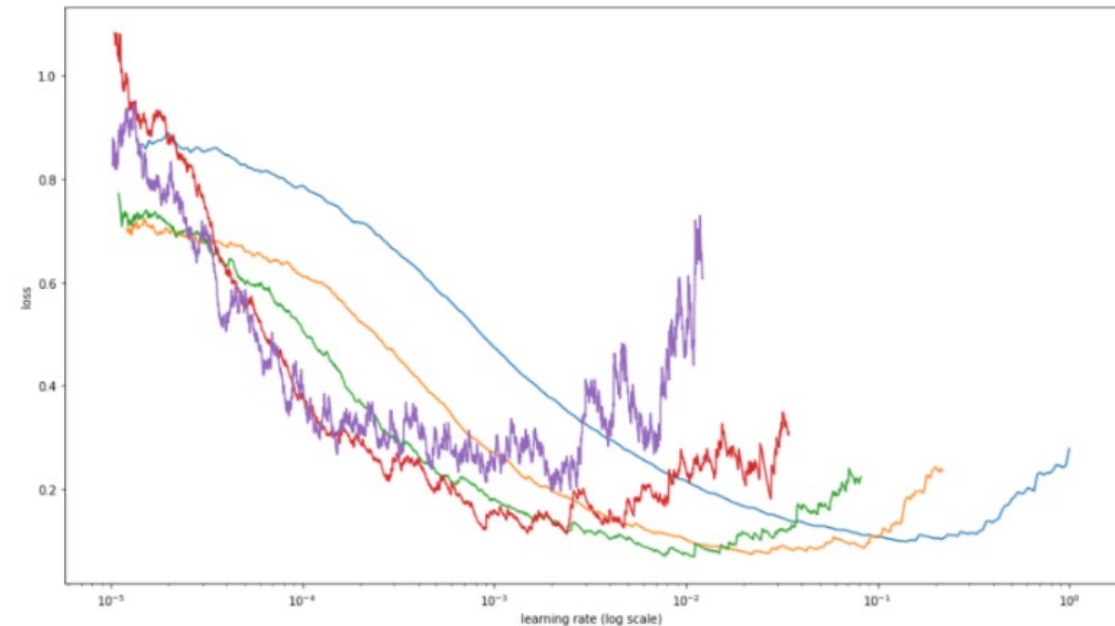
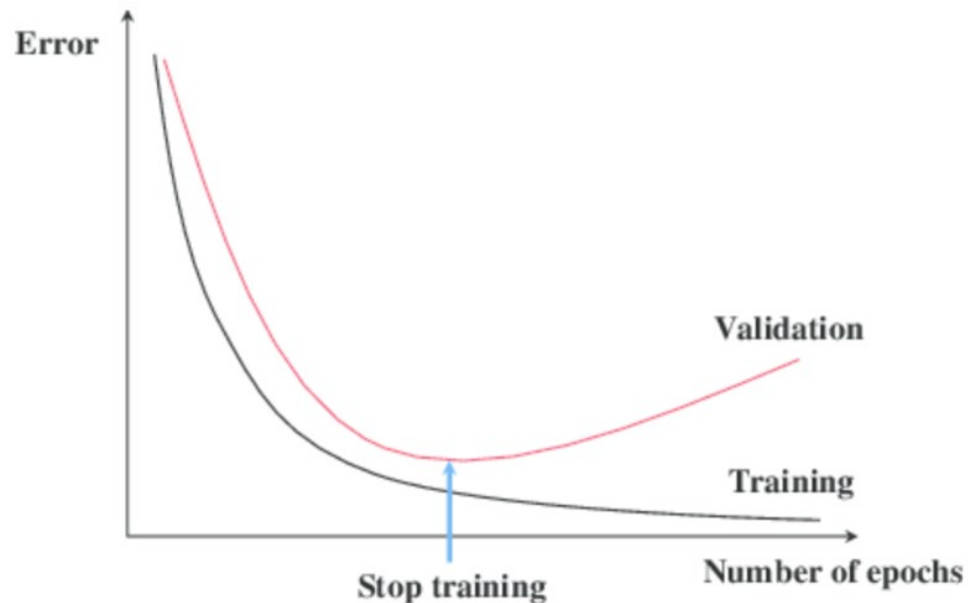
- ❖ Too little time = underfit
- ❖ Too much time = overfit

- **Solution 1:**

- To stop training as soon as the validation error reaches a minimum.

- **Solution 2** (for Stochastic and Mini-batch Gradient Descent):

- To stop only after the validation error has been above the minimum for some time (when you are confident that the model will not do any better).



Weekly Report 13

08/17/2023

UNH
ECE

DEPT.

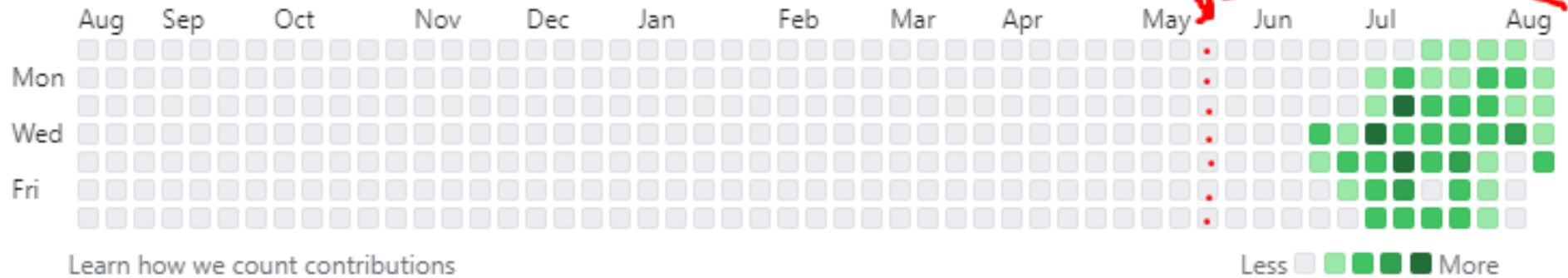
The progress of my project

1

159 contributions in the last year

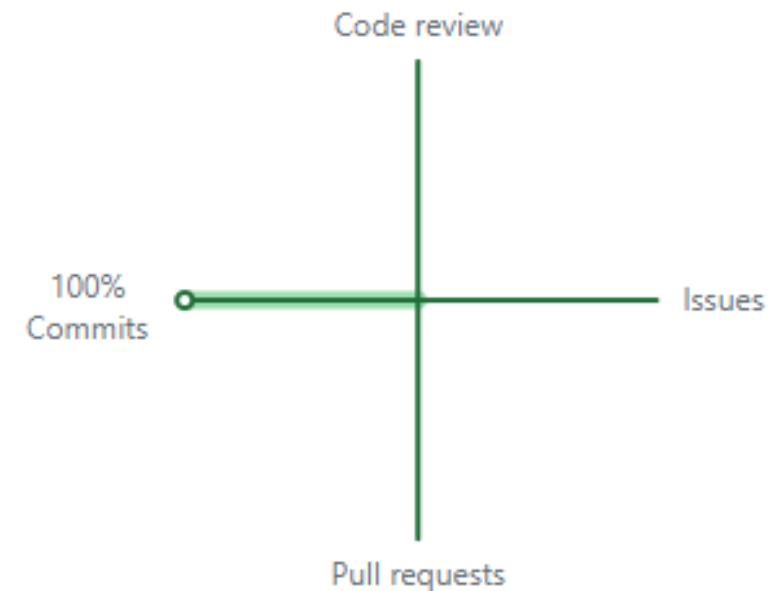
Contribution settings ▾

2023



Activity overview

Contributed to [Alireza-Motazedian/GIC](#),
[Alireza-Motazedian/Alireza-Mota...](#),
[Alireza-Motazedian/Teaching_Pyt...](#)
and 5 other repositories





main 1 branch 0 tags



Alireza-Motazedian Update Daily logs.md



0. Archive



1. Report



2. Code



3. Data



4. Concepts _ Books _ Papers



.gitignore



README.md



requirements.txt

Folder PATH listing

```
.
|
+---0. Archive
|   Contract Summer 2023.pdf
|   Links _ Resources _ People.md
|   Proposal-GIC-Power Grid.pdf
|
+---1. Report
|   Daily logs.md
|   Report Slides.pptx
|
+---2. Code
|   Data provided by Talha.ipynb
|   GIC Data Exploration.ipynb
|
+---3. Data
|   |   filled_median_1995-2011_2015.parq.gzip
|   |   filled_median_engineered_normalized_1995-2011_2015.parq.gzip
|   |   merged_1995-2011_2015_Wierd_Values_replaced_with_NAs.parq.gzip
|   |   merged_1995-2011_2015.parq.gzip
|   |   solar_wind_1995-2022.parq.gzip
|   |   SuperMAG_1978-2021.parq.gzip
|   |   README.md
|
+---images_GIC_data_download
|   1.png
|   2.png
|   3.png
|   4.png
|   5.png
|   6.png
|   7.png
```

\---4. Concepts _ Books _ Papers

```
|   README.md
|
+---Books
|   01. Machine Learning Techniques for Space Weather.pdf
|
+---images_GIC
|   Figure_0.png
|   Figure_01.png
|   Figure_02.png
|   Figure_03.png
|   Figure_04.png
|   Figure_05.png
|   Figure_06.png
|   Figure_07.png
|   Figure_08.png
|   Figure_09.png
|   Figure_10.png
|   Figure_11.png
|   Figure_12.png
|   Figure_13.png
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|   Figure_33.png
|   Figure_34.png
|   Figure_35.png
|   Figure_36.png
|   Figure_37.png
|   Figure_38.png
|   Figure_39.png
```



GIC

Private



main ▾



1 branch



0 tags



Alireza-Motazedian Update Daily logs.md



0. Archive



1. Report



2. Code



3. Data



4. Concepts _ Books _ Papers



.gitignore



README.md



requirements.txt

\---Papers

1. [Link](#) Keesee et al. 2022.pdf
2. [Link](#) Pinto et al. 2022.pdf
3. [Link](#) Physics-Informed Machine Learning A Survey.pdf
4. [Link](#) (IEEE) Analysis of Magnetometer Data from a Strong G3 Geomagnetic Disturbance.pdf
5. [Link](#) (IEEE) Real-Time Monitoring Applications for the Power Grid under Geomagnetic Disturbances.pdf
6. [Link](#) Sarker 2021.pdf
7. [Link](#) Siddique et al. 2022.pdf
8. [Link](#) (IEEE) Modelling considerations for the Hydro One real-time GMD management system.pdf
9. [Link](#) (IEEE) A Machine Learning Approach to Detection of Geomagnetically Induced Currents in Power Grids.pdf
10. [Link](#) (IEEE) A Review of Geomagnetically Induced Current Effects on Electrical Power System Principles and Theory.pdf
11. [Link](#) (IEEE) Deep Learning-Based Channel Prediction for Edge Co.pdf
12. [Link](#) Impact of geomagnetic disturbances on power transformers.pdf
13. [Link](#) Insight into impact of geomagnetically induced currents on power.pdf
14. [Link](#) (IEEE) Optimal Placement of GIC Blocking Devices Considering Equipment Thermal Limits and Power System Operation Constraints.pdf
15. [Link](#) Global Geomagnetic Perturbation Forecasting Using Deep Learning.pdf
16. [Link](#) The Role of GlobalRegional Earth Conductivity Models in Natural Geomagnetic Hazard Mitigation.pdf
17. [Link](#) Analysis of Geoeffective Impulsive Events on the Sun During the First Half of Solar Cycle 24.pdf
18. [Link](#) Measurement of geomagnetically induced current (GIC) around Tokyo, Japan.pdf
19. [Link](#) Towards an AI-based understanding of the solar wind A critical data analysis of ACE data.pdf
20. [Link](#) A Novel Algorithm for Imputing the Missing Values in Incomplete Datasets.pdf
21. [Link](#) Gap filling of solar wind data by singular spectrum analysis
22. [Link](#) A Review of Missing Values Handling Methods on Time-Series Data
23. [Link](#) Logistic Regression with Missing Data: A Comparison of Handling Methods, and Effects of Percent Missing Values
24. [Link](#) Evaluation of three Simple Imputation Methods for Enhancing Preprocessing of Data with Missing Values

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 Code  Issues  Pull requests

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Daily logs (Summer 2023)

- 8 h/day
 - Every hour:
 - 45 minutes: Work
 - 15 min: Break (Meditation, Drinking water, Walking, Nap, Workout)
- Week 13 _ August 14 - 18
 - 08-18-2023 Friday
 - 09-10:
 - 10-11:
 - 11-12:
 - 12-01:
 - 01-02:
 - 02-03:
 - 03-04:
 - 04-05:
 - 08-17-2023 Thursday
 - 09-10:
 - 10-11:
 - 11-12:
 - 12-01:
 - 01-02:
 - 02-03:
 - 03-04:
 - 04-05:

Roadmap on our paper

Table: From later listed to supervised. A Comparative Implication of Missing Data Filling Methods: Learning Machine and Deep Learning Algorithms.

1. Introduction

Background

Problem Statement

Objectives

Related Work

Roadmap on our paper

Methods

Dataset Description

Missing Data Imputation Techniques

Machine/Deep Learning Models

Roadmap on our paper

Experiments & Results

Experimental Setup

Evaluation Metrics

Results

Discussion

Roadmap on our paper

Conclusion & Future Work

Proposed Directions for Future Research

Acknowledgments & References

Heartful Thanks

Offering me the opportunity to join your research group

Allowing me to work with a real dataset

Under your mentorship, I've gained a profound understanding of the core concepts

Your guidance during our meetings has been instrumental to my progress

I'm also thankful for the support from the other group members

I'm keen to collaborate further with your research group in the future

Thank You

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 GIC Data Exploration.ipynb

 3. Data

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 1. Libraries & paths & functions

 1.1. Liberraries

 1.2. Paths

 1.3. Functions

 2. Data acquisition method

 2.1. Solar wind data (OMNIWeb)

 2.2. SuperMAG data

 3. Data acquisition and compression

 3.1. Solar wind data

 3.1.1. Description

 3.1.2. Code (Do not run it!)

 3.2. SuperMAG data

 3.2.1. All years

 3.2.1.1. Description

 3.2.1.2. Code

 3.2.2. Year by year (1995)

 3.2.2.1. Description

 3.2.2.2. Code

 3.2.3. all_years vs. year_by_year

4. Merge datasets

4.1. Solar wind data

4.2. SuperMAG data

4.3. Merged (Solar wind & SuperMAG)

5. Data exploration

5.1. Description

5.2. Data types

5.3. Dealing with NAs

5.3.1. Missing values exploration

5.3.2. Weired values to NAs

5.3.3. Percentage of missing data

5.3.4. Missing values treatment

6. Feature Engineering (dBH)

7. Normalization

8. Plots

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5.3. Dealing with NAs

5.3.1. Missing values exploration

5.4. values exploration

5.4.1. Statistics

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Note:

- Due to their large size, the datasets have not been uploaded to this repository. Below are links to our Google Drive, where you can download the datasets.
- Alternatively, using the following tutorial, you can download the data you need from the respective websites.

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- [1. Links to our data](#)
- [2. Data acquisition method](#)
 - [2.1. Solar wind data \(OMNIWeb\)](#)
 - [2.2. SuperMAG data](#)
- [3. Datasets descriptions](#)
 - [3.1. Brief descriptions](#)
 - [3.1.1. Solar wind](#)
 - [3.2. Full descriptions](#)

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 - 2.2. Geomagnetic Induced Currents (GIC)
 - 2.3. Concept of UT
 - 2.4. Dst index
 - 2.5. Normalization
- 3. Website
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 - 3.1.1. Magnetometer
 - 3.1.2. Solar Wind Electron Proton Alpha Monitor
 - 3.2. SuperMAG
 - 3.2.1 SuperMAG definition
 - 3.2.2. Subtract Baseline
 - 3.2.3. High Fidelity and Low Fidelity
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 - 4.2. SuperMAG (description)
 - 4.3. The magnitude of variables
 - 4.4. Feature scaling vs. normalization
- 5. Machine Learning
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 - 5.3. ADAM
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 - 5.4.1. Mean
 - 5.4.2. Median
 - 5.4.3. Mode
 - 5.4.4. Constant number or string
 - 5.4.5. Miss Forest
 - 5.4.6. Mice Forest
 - 5.4.7. KNN Imputation
 - 5.4.8. Bayesian imputation
 - 5.4.9. Multiple imputation
 - 5.4.10. Expectation-maximization
 - 5.4.11. stochastic regression imputation
 - 5.5. Principal component analysis (PCA)
 - 5.6. LSTM
 - 5.7. CNN
 - 5.8. SVM
 - 5.9. RF
- 6. Papers
 - 6.1. PILM: A Survey on Problems, Methods and Applications
 - 6.2. a real-time GMD monitoring system
- 7. Questions
 - 7.1.1. Question 1
 - 7.1.2. Question 2
- 8. Code optimization
 - 8.1. Pure Python to Scientific Python

Title: From Solar Wind to SuperMAG: A Comparative Exploration of Missing Data Filling Methods Leveraging Machine and Deep Learning Algorithms

1. Introduction

➤ **Background:**

- The significance of predicting solar storms.
- The challenges of dealing with noisy time-series data.

➤ **Problem Statement:**

- Highlight the issue of missing data in solar wind datasets.
- The necessity of choosing an appropriate method to handle missing values, as it can significantly influence the quality of predictions.

➤ **Objective:**

- The aim of the paper is to compare different missing data imputation methods
- Evaluate their performance using various machine learning models.

2. Related Work

- Review previous works related to solar wind data analysis.
- Discuss previous methods employed to handle missing data in this or related fields.
- Identify gaps in the existing literature (e.g., lack of comprehensive comparison, outdated imputation techniques, etc.)

3. Methods

➤ Dataset Description:

- Describe the dataset in detail, highlighting the features and their significance.
- Provide statistics on the missing data.

➤ Missing Data Imputation Techniques:

- Conventional Methods: (Mean and Mode imputation, and deletion)
- Estimation Methods:
 - ✓ Miss Forest
 - ✓ Mice Forest
 - ✓ KNN Imputation
 - ✓ Bayesian imputation
 - ✓ Multiple imputation
 - ✓ Expectation-maximization
 - ✓ Stochastic regression imputation

➤ Machine /Deep Learning Models:

- ✓ LSTM
- ✓ CNN
- ✓ SVM
- ✓ RF

4. Experiments & Results

➤ Experimental Setup:

- Discuss how you split the data (train/test/validation).
- Describe any preprocessing steps (aside from the imputation).

➤ Evaluation Metrics:

- Explain the metrics you'll use to evaluate the performance (e.g., RMSE, MAE, etc.).

➤ Results:

- Tabulate or visualize the results for each combination of the imputation method and machine learning model.
- Highlight which imputation method works best for each model.

5. Discussion

- Analyze the results in-depth.
 - Why did some methods perform better than others?
 - Are there any patterns?
- Discuss the real-world implications of your findings, especially in the context of predicting solar storms and mitigating their impacts.

6. Conclusion & Future Work

- Summarize your findings.
- Propose directions for future research.
 - Improving the imputation methods
 - Investigating other machine learning models
 - Applying the methodology to other datasets.

7. Acknowledgments & References

- Thank those who contributed to the paper or provided data/resources.
- List all references you cited in the paper.

- ✓ Offering me the opportunity to join your research group.
- ✓ Allowing me to work with a real dataset.
- ✓ Under your mentorship, I've gained a profound understanding of the core concepts.
- ✓ Your guidance during our meetings has been instrumental to my progress.
- ✓ I'm also thankful for the support from the other group members.
- ✓ I'm keen to collaborate further with your research group in the future.

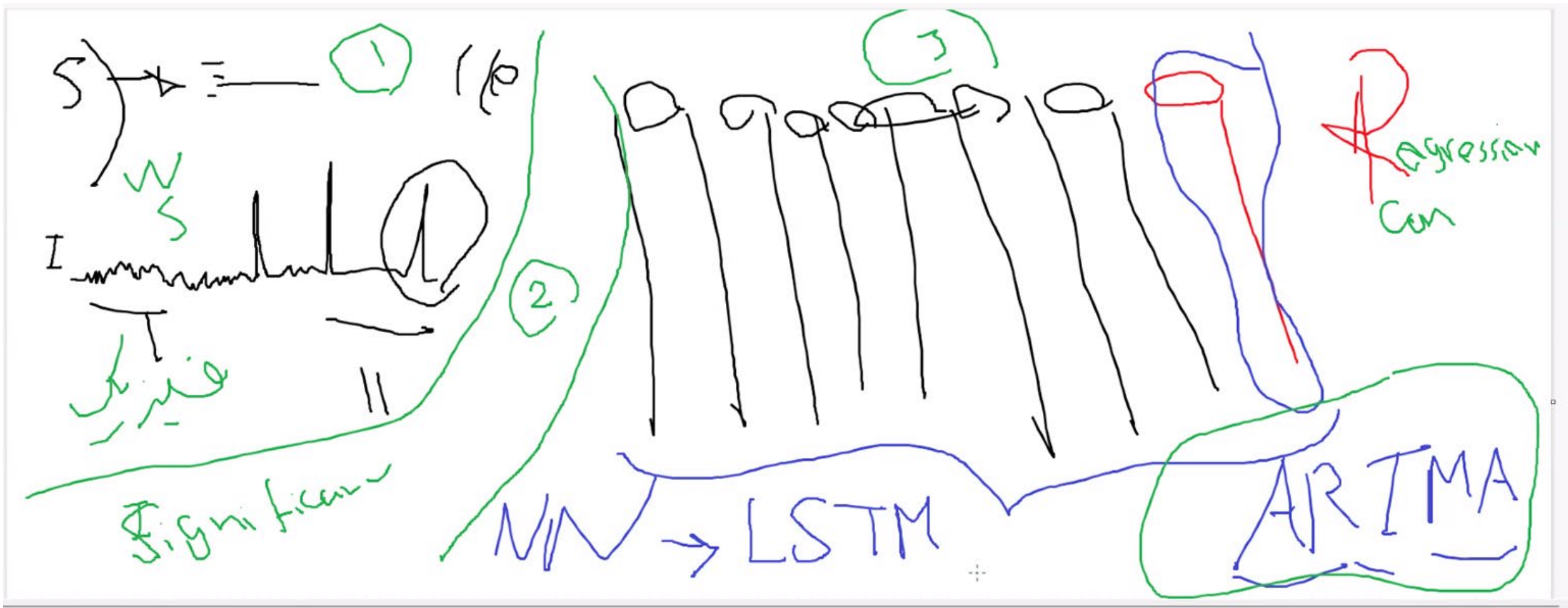


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Learning : n unique - unique - value count

Ex) class

No.	grade	n unique	unique	value count	n unique	unique	value count	n unique	unique	value count	n unique	unique	value count
1	A	4	A	3	4	3	1	1	1	4	1	1	1
2	A		B	4		4	1						
3	A		C	1		1	1						
4	B		D	2		2	1						
5	B												
6	B												
7	B												
8	C												
9	D												
10	D												



Missing streak exploration

4

NA: 97k

↓
Streak: 8k

- max length: 9k
- Average: 12
- min: 3
- median: 7
- n-unique: 120

length How many?

Exn 10 → 70 = 700

20 → 90 = 1800

120

8k

unique: 16

Missing-Streaks exploration

NA: 97333

↓

Streak: 7803

- Max length: 9754
- Average length: 12
- min length: 3
- median length: 7
- n-unique: 120

unique length	number of streak
0	7
1	6
2	8
3	5
4	4
5	1
6	1
7	1
8	1
9	1
10	1
11	1
12	1
13	1
14	1
15	1
16	1
17	1
18	1
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113	1
114	1
115	1
116	1
117	1
118	1
119	1
120	1

number of unique value count of lengths of missing-streak

16 state

97333 NA